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THE VARIABLE TRAJECTORY: A SYSTEMATIC REVIEW, ANALYTIC SYNTHESIS AND CONSTRUCT DOMAIN CONSOLIDATION OF INTERNATIONAL MEASURES OF COMPETENCE IN DOCTORS AND MEDICAL STUDENTS

Running title: Measuring competence in doctors

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The variable trajectory:
A systematic review, analytic synthesis and construct domain consolidation of international measures of competence in doctors and medical students

ABSTRACT

Background

Competence is assessed throughout a doctor's career. Failure to identify and manage impaired competence can have critical consequences. Consistent conceptualisation and accurate measurement of this construct is imperative. Therefore, the objective of this review was to identify and evaluate measures used to assess competence in doctors and medical students.

Methods

A systematic search of the published literature was undertaken between December 2019 and February 2020 for articles reporting on the measurement of competence in doctors and/or medical students. Searches were conducted in the PsychSOURCE, US National Library of Medicine National Institutes of Health, MEDLINE (PubMed), The Cochrane Central Register of Controlled Trials (CENTRAL), and Web of Science electronic databases. Citation screening and forward citation tracking of included studies was carried out to identify any further relevant papers for inclusion. One thousand one-hundred and thirty-six potentially relevant articles were screened. An analytic synthesis approach was implemented to the identification, organization and interpretation of homogenous study and measure characteristics.

Results

Twelve competence domains were identified from the 153 identified measures. Knowledge and Procedural competence domains were the dominant focus of publications reporting current medical practice, but less so in research-based studies which more frequently assessed interpersonal, psychological, physiological and ethical competencies. In the 105 included articles,

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the reporting of measurement instrument quality was varied, with comprehensive reporting only present in 53.6% of measures; validation for some of the measures was particularly limited.

Discussion

Whilst this review included a considerable number of publications reporting the measurement of competence in doctors and medical students, the heterogeneity of the measures and variation of findings limit the ability to evaluate their validity and generalisability. However, this review presents a resource for researchers and medical educators which may inform operational practice and future research.

Keywords: Competence · Doctor · Medical student · Physician · Performance

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Strengths and limitations of this review

- To our knowledge, this is the first contemporary systematic review to consolidate globally recognised domains relevant to competence assessment of doctors and medical students in both research and in-practice measures.

- This PROSPERO registered review employed an extensive and rigorous search strategy of four databases and forward citation searches using defined selection and abstraction criteria calibrated by three researchers, undertaken in alignment with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement, and guided by the synthesis without meta-analysis (SWiM) reporting guidelines.

- The review identified a broad range of measures used to assess competence, and identifies the homogeneity and heterogeneities using a textual and conceptual analytic synthesis approach.

- A limitation of this review is that English language restrictions were placed on the searches, and although research studies were derived from 18 countries, the inclusion of further languages may yield unidentified studies, and subsequent differences in worldwide assessments of competence.

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INTRODUCTION

The timely identification and support of doctors who do not demonstrate sufficient competence to undertake their duties safely is imperative. However, the proportion of doctors who are incompetent, or dyscompetent, failing to uphold adequate practice standards is unclear[1]. Robust assessment of competence necessitates accurate and standardised measures that comprehensively examine all domains relevant to the core competencies required to undertake a job effectively. It has been argued that competencies are learned, complex context-specific dispositions, but the complexity and diversity inherent in doctors' responsibilities make measurement of competence a complex task[2].

Despite numerous definitions,[3] an agreed explicit, amalgamated, and quantifiable definition of the construct of competence, and in many cases a consistent and transparent chain of inferences to validate the observed assessment and measurement of competence are lacking[4]. It has been argued,[5] that it is challenging to identify reliable and valid measures of competence as many of the skills are considered difficult to quantify. Contemporary competence-based medical education is predominantly driven by patient safety, efficiency and clinical outcomes[6].

In order to explore these challenges, and provide an overview of the diverse facets of competence, and their dominance within current assessment measures; the objective of this systematic review of peer-reviewed scientific literature was to identify and evaluate specific measures of competence relevant to doctors and medical students.

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METHOD

The review protocol was registered with PROSPERO, adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement,[7] and guided by the 'synthesis without meta-analysis (SWiM) reporting guidelines'[8].

Literature Search

Data Sources

A systematic search of the literature was conducted in the PsychSOURCE, US National Library of Medicine National Institutes of Health, MEDLINE (PubMed), The Cochrane Central Register of Controlled Trials (CENTRAL), and Web of Science electronic journal databases. Citation screening and forward citation tracking of included studies was carried out to identify any further relevant papers; these were retrieved from the journal site or academic library databases.

Search Strategy

A pilot search was undertaken to determine relevant electronic search engines, refine the search strategy, define search keywords, identify relevant MeSH terms, determine search restrictions, and select Boolean operators. English and American-English language terms were used. Search terms employed the keywords; 'doctor', 'competence', and 'performance', incorporating the Boolean operator ('and'), in addition to MeSH terms (physicians, medical students). Restrictions on the databases included human studies, published in English language, and full text availability. No date restrictions were applied. Searches were undertaken between December 2019 - February 2020. Further searches in response to the literature were performed through reference lists of reviewed articles in forward citation searches. Results were downloaded to reference management software and de-duplicated.

Study Identification and Selection

Inclusion/Exclusion Criteria

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Studies were included if they described the measurement of competence in medical students and doctors, with full-text availability in peer-reviewed journals, published in English language. Commentaries, editorials, letters, book chapters, conference papers, books, doctoral theses, dissertations, and articles which did not adequately report competence assessment measures were excluded.

Calibration of Findings

One reviewer undertook title and abstract screening, and selection methods were supervised by two reviewers who verified the extraction of data, checked consistency and methodological clarity. Study selection disagreements and subsequent queries about inclusion following full-text review were resolved by discussion between researchers to reach consensus.

Study Selection

From searches, 1,136 papers were identified; after duplicates were removed 648 original articles remained. Following assessment for relevance to the objectives and selection criteria 105 articles were selected for systematic analysis. **Figure 1** reports the PRISMA (2009) flow diagram of study selection, and specific search engine results are presented in *supplementary file 1*.

Data Extraction

Selected articles were subjected to three preliminary levels of screening: Title, Abstract and Full Text. The following data (when available) from articles was extracted using a pre-designed and piloted data-extraction pro-forma which included the following categories and sub-categories: (1) Referencing information, (2) Quality assessment, (3) Study characteristics (geographical origin, sample, objectives, design), (4) Type of measure, (5) Assessment to examine whether the properties of each measurement tool (construct validity, content validity, criterion validity, internal consistency, test-retest reliability, inter-rater reliability etc.) had been reported, (6) Main results/outcomes, (7) Additional further information relevant to the objectives. Results were recorded in an Excel spreadsheet and each study was assigned a reference identifier code (*supplementary file 2*).

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Study Assessment of Methodological Quality

Quality inclusion criteria were restricted by whether sufficient detail was provided to enable analysis of methodological quality and the evaluation of study measures. The five 'appraisal prompts for informing judgements about quality of papers' defined by Critical Interpretive Synthesis (CIS)[9] methodology were evaluated and tabulated for all included studies. CIS yielded quality appraisal nominal data scores for each question (1 yes, 0 no), with a cumulative score ranging from 0 to 5; scores ranged from 2 to 5. Ninety-four of the 105 studies achieved the maximum (5/5) cumulative rating score, six scoring (4/5), four scoring (3/5), and one scoring (2/5).

Data evaluation and synthesis

As the protocol anticipated, due to the heterogeneity of measurement tools in this review, and inconsistencies in methodology and reporting of results, a meta-analysis was not possible. Preliminary categorisation of the studies included data extraction of study characteristics, types of competence measures and measurement properties. A conceptual analytic synthesis of identified dimensions of similarity and difference in defining the construct of competence in this population was undertaken.

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RESULTS

The search strategy generated 105 articles selected for full data abstraction: 13 from PsychSOURCE, 19 MEDLINE/PubMed, 24 Cochrane Library – CENTRAL, 21 Web of Science, and 28 from specific paper retrieval forward citation searches at the journal site. Selected reviewed papers in the results section are referenced in the format[P:1,2,3,4,5], and individual measures are referred to as[M:1,2,3,4,5] (see *supplementary file 2* for assigned reference identifier codes and corresponding references).

Characteristics of Selected Studies

Included articles were published between 1990 and 2020. There was a progressive increase in the number of publications per year over this period (mean annual publications per year 0.3 from 1990 to 1999, 1.9 from 2000 to 2009, and 8.2 from 2010 to 2019), consistent with increasing research interest and societal awareness of the extent and consequences of medical error[P105]. Studies were restricted to English language, but without restriction on geographic origin and were derived from 18 countries, the majority from the USA (n = 34), UK (n = 17), Germany (n = 15), and Canada (n = 15). The study designs included: two intervention, 22 measurement development, 26 measurement evaluation, 17 curriculum development, 40 clinical assessment, 21 training evaluation, and 11 experiential assessment; 26 studies comprised multiple categories.

Competence Assessment Measures

From the 105 articles, 153 measures of competence were identified, 98 of which were unique to that study (a comprehensive list of all measures is presented in *supplementary data file 3*). The competence assessment measures were categorised as either established 'in-practice measures' (IM) currently implemented for the assessment of doctors and medical students, or 'research-measures' (RM) developed or applied to assess competence for research evaluation, but not at the time of publication employed for occupational competence assessment. Thirty-eight studies examined IM, 45 examined RM, and 22 explored both RM and IM. Most (n = 76) of the selected studies were single-institution assessments - limiting the generalisability of findings, 25 were multicentric (based at multiple institutions), and four were not institution-based.

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Measures were categorised by their procedural comparability, and seven dominant operational characteristics were identified: (1) Clinical Performance Assessment - was applied in 30.7% of measures (n = 38 IM, n = 9 RM), typically assessed using simulated patient ratings to assess clinical competencies such as communication skills and diagnostic reasoning in response to presented symptoms; (2) Anatomic Model Simulation – providing simulations of human anatomy in order to test knowledge and used in 6.5% of measures (n = 9 IM, n = 1 RM); (3) Psychometric Assessment – predominantly evaluated personality, aptitude, reasoning and emotionality, and conducted in 7.8% of measures (n = 2 IM, n = 10 RM); (4) Patient Records - to assess the medical record keeping accuracy and completeness, were utilised in 1.3% (n = 2 IM); (5) Computerized Testing – included a variety of computerised assessments, used in 9.8% of measures (n = 1 IM, n = 14 RM); and (6) Written Testing – included multiple-choice and short-answer questions in traditional pen-and-paper testing in 31.4% of measures (n = 19 IM, n = 29 RM). The prevalence of each measurement category demonstrated a preference for the application of written testing and clinical performance assessment in the assessment of competence. Fifty-six (36.6%) measures applied a unidimensional procedure to assess the target domains, and 97 (63.4%) operated multidimensional procedures.

Established In-practice Measures

Competence was assessed through a variety of IM methods. Objective structured clinical examinations (OSCEs) typically involve patients or simulated patients and the most duplicated measure (in 20 articles). OSCEs are frequently considered[P:53] the ‘gold standard’ for assessing competence skills, as they are high-fidelity, can be reliable, and are regarded as valid because of their ability to differentiate between occupational levels of aptitude and specialisms. The use of standardised patients (SPs) endeavours to replicate doctor-patient consultations can identify variation in clinical practice between doctors in a realistic simulated setting. SP single-case assessments were used to measure specific competencies in nine of the studies.

Anatomic model simulation testing, employed in 10 measures, is widely used as a safe method to develop procedural competence in early medical education. However, whether the

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anatomic-model patients are the most valid means of assessing procedural competence is widely debated[P38]. Workplace-based assessment (WBA) such as mini-clinical evaluations (Mini-CEX) and direct observation of procedural skills (DOPS) are heavily integrated in medical education. However, the Mini-CEX and DOPS lack standardisation, cannot be used with large numbers of participants simultaneously, are costly, time consuming, and transferability of skills is not always guaranteed in a simulated assessment setting; particularly in a workplace with high levels of social, psychological and physiological stress[P:20]. Four studies[P:20,29,79,89] examined 360-degree assessments (i.e. rating evaluations from colleagues within the workplace)

Written tests provide reliable assessment of several domains, particularly knowledge and psychometric assessment. Three studies[P:2,95,97] used the Jefferson Scale of Physician Empathy (JSE), one [P:3] employed the Jefferson Scale of Physician Empathy – Student Version (JSPF-S). Written tests measuring knowledge included two studies using the US Medical Licensing Examinations (USMLE), and three studies that included the Medical College Assessment Test (MCAT) - an assessment used by many medical schools as part of the entry-selection process. However, written tests are considered insufficient to measure behavioural clinical competence [P:20]; therefore, there has been a traditional reliance on the judgement of competence by educators or seniors in demonstration-observation tests of clinical performance skills. The combination of a knowledge test (written assessment) with an applied test of clinical skill (OSCE) was shown to increase predictive validity versus either alone[P:12]. Vignettes (five studies) and written case simulations have been widely used to assess competence in both clinical training and experimental research, because of their ease of administration, negligible cost, and objective quantifiable results. However, they do not assess important social interaction skills such as history taking, nor assess competence in a real-world situation including occupational stressors. Amongst IM measures there was significant variability in testing procedures, content domains and inconsistent assessment outcomes, therefore these were not grouped for evaluation purposes.

Contemporary Experimental Research Measures

Electronic-technology has an increasing role in healthcare, but raises new challenges for competence assessment; one study presented an electronic platform[M:149] that automatically

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assesses a doctor's competence from online textual consultations with patients utilizing a novel machine that provides auto-evaluation based on patient satisfaction following online consultation. In addition to patient evaluations, innovative simulation-based medical education (SBME) and assessment provides alternatives to practice and assessment without risk to patients[P:26].

In an era where doctors' retirement age has extended, further cognitive competence considerations have emerged. Forty-two measures in 37 of the studies assessed cognitive competence, publication dates indicating that research interest in this area has markedly risen in recent years (7.6% of the studies between 2000 and 2009, compared to 26.7% from 2010 to 2020). Neuropsychological screening found[P:81] that many of the doctors who fell significantly below expected competence assessment levels had cognitive impairment sufficient to explain their diminished competence. Another study developed 'the Mini-NeuroCart'[M:80], a cognitive and psychomotor test battery for assessment of subjective and objective measures of alertness, mood, concentration and self-assessed procedural competence, and showed that the competence of postcall surgeons was similar to, or worse than that of ethanol-intoxicated surgeons.

Competence Domains

Studies were categorised according to the competence assessment domains. Twelve ascendant domain categories based upon their prevalence in the selected measures were identified, with descriptors of these in **table 1**.

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Table 1 Domain Descriptors in Descendent Order of Prevalence in All Measures

Number	Domain	Prevalence	Descriptor
1	KNOWLEDGE	55.5%	The acquisition and retention of learning.
2	PROCEDURAL COMPETENCE	45.1%	The ability to perform specific technical skills/procedures required for the occupational obligations.
3	JUDGEMENT & BIAS	41.8%	An ability to nullify or negate judgement biases to competently work in the benefit of patients and colleagues.
4	COMMUNICATION	40.5%	The ability to engage patients and colleagues in effective interaction.
5	PROFESSIONALISM	29.4%	To uphold the expected traditional occupational behavioural standards of the profession, and regulatory code of conduct.
6	COGNITIVE COMPETENCE	27.4%	To effectively utilize the cognitive resources required for the selection and application of skills.
7	ENVIRONMENTAL COMPETENCE	23.5%	This domain relates to the ability to maintain acuity in environmental adversity.
8	COPING COMPETENCIES	21.5%	Competence to psychologically manage the complex occupational exigencies.
9	SELF-ASSESSMENT ACCURACY	20.9%	Self-assessment and accurate self efficacy in continuous self-monitoring of competence.
10	EMPATHIC COMPETENCE	20.2%	Empathic Competence describes the awareness and responsiveness to patients' and colleagues' feelings and experiences.
11	ETHICS & PATIENT SAFETY	16.9%	A commitment to exemplar medical practice by prioritising patient safety and ethical standards.
12	PHYSIOLOGICAL COMPETENCE	6.5%	Physiological competence describes a state of optimal functioning that supports the physical and psychological workplace demands.

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Consideration of Each Domain Within the Literature

Knowledge (**Domain 1**), deemed as the primary proximal determinant of competence[P:11] was the most commonly assessed competency in IM, historically assessed through written examinations[P:31], contemporary measures endeavour to combine knowledge with procedural assessment. Concordantly, only 27.2% of measures conformed to traditional written-test approaches, though 6.4% of computerised test measures were a comparable format.

Within **Domain 2** (Procedural Competence), 54.6% of the measures employed a clinical performance test; other measures included patient vignettes, written or computerised tests (30.2%), and anatomic model simulation (15.6%).

Domain 3 (Judgement and Bias) was addressed in 41.8% of measures and incorporated competence with diversity, and awareness of bias in an occupational context. Cultural competence was found to be lacking in one study[P:8] as a result of an interaction between race and treatment outcomes as there was a higher likelihood of doctors escalating the use of opioids for African American patients. Sexual and gender minority individuals experience high rates of discrimination when seeking healthcare, contributing to patient care disparities[P:68]. Furthermore, doctors reported discomfort when treating people who have disabilities[P:96], as assessed by the Scale of Attitudes Toward Disabled Persons (SADP)[M141].

Domain 4 (Communication): Competent doctor-patient and inter-professional communication is included within this domain. Ineffective communication with colleagues has been associated with reduced patient safety and risk to care quality[P:71], and the most frequent cause of complaints against doctors is related to poor communication[P:42].

Though doctors are expected to uphold the customary standards of professionalism (**Domain 5**), there was no consensus on its definition[P:77], and an absence of explicit standardised factors measured within this domain[P:78]. Most (73.8%) of the measures used subjective rating scales; many within OSCE and WBA. Remarkably, although 36 IM measured professionalism, only 9 RM assessed this domain.

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Cognitive competence (**Domain 6**) denotes the specific cognitive resources required for the selection and application of skills such as diagnostic and clinical reasoning. Wechsler Adult Intelligence Scale (WAIS-IV) assessments have found cognitive resource differences between doctors who experience difficulties, and those who do not[P:70]. The cognitive concept of 'mental workload' perceives the brain having a limited capacity to process stimuli, research has found that high levels of workload are associated with error and poor competence[P:19]. Traditional measurement of task performance does not necessarily predict competence when a combination of tasks must be undertaken under time pressure resulting in cognitive overload, and the 'task-shedding' of key competencies resulting in risk of impaired competence.

Environmental Competence (**Domain 7**) employs measures that account for (or simulate) the context of occupational pressures in assimilating and responding to environmental context[P:72] using real-time information management[P:54]. One study[P:19] examined the effect of multitasking on competence in medical students using a secondary task method employing a venepuncture procedure[M:32] alongside an electronic application[M:31] which demanded a tapping response when the device vibrated, mental workload was measured in time delay and effects were observed on communication deterioration. Measures that included time (chronometry)[P:15,19,26,34,46] highlighted that timed tests increased the reliability and validity of measuring diagnostic accuracy[P:15], as time-pressure has been found to disrupt the dynamic interaction between reasoning systems influencing diagnostic tasks[P46]. Conclusions about competence may be flawed unless contextual environmental factors are considered[P:72].

Doctors are expected to handle limited resources, pressure, and life or death performance outcomes that often carry an emotional load; 'Coping Competence' (**Domain 8**) is therefore paramount[P:72]. Distress and inability to cope has been associated with medical errors[P:105]. Therefore, the ability to cope with environmental, cognitive, physical and emotional workplace demands is key to competence and occupational fulfillment[P:104].

The ability to accurately 'self-assess' and work at a level of occupational aptitude. (**Domain 9**) is key to the identification of difficulties, and safe practice. RM focussed more heavily on self report measures (n = 27) than IM (n = 5). Concerningly, one study[P:3] found that self-rated empathy and observer ratings were not associated, and another[P:7] found no

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correlation between confidence scores and clinical or knowledge abilities. A doctor's ability at any career stage to accurately self-assess their competence acquisition provides assurance of working within scope of practice[P:14].

Medical educators and professional regulators recognise the importance of Empathic Competence (**Domain 10**); strong associations were found between observer ratings of empathy and OSCE scores[P:3]. The prominence of empathic competence within the selected measures was comparable in RM (n = 16) and IM (n = 15). In medical education research[P:95], empathic competence has been demonstrated to increase in male trainees and decrease in females during the course of training. Importantly, empathic competence has been found to be strongly associated with patient satisfaction[P:97].

Domain 11 relates to adherence to ethical standards and commitment to patient safety. This domain was addressed in regulatory assessments[e.g.M:129,130,138], rating scales assessed by senior doctors[e.g.M:1,9,91,117,126], WBA's[e.g.M:146,147], patient evaluations[M:36], and values-based measures such as a questionnaire survey about notions of a 'good doctor'[M:8]. Neuropsychological testing[M:123] has been found to be a strong predictor of risk to competence[P:86]; a consistent decline in competence has also been identified in recertification exams in America related to years since graduation[P:82]. Another study[P:75] developed and tested an OSCE designed to directly assess socio-cultural dimensions of patient safety competency[M:117], demonstrating sufficiently reliable station scores in this domain. Engagement with continuous competence improvement and collaboration with colleagues (rather than isolated practice) has been identified as a protective factor[P:28].

Physiological competence (**Domain 12**) emerged as a domain rarely considered in assessment. RM measures within this domain included psychometric measures of burnout[M:18,52], the Epworth Sleepiness Scale[M:151], and the Medical Outcomes Study Short Form (SF8) psychological and physical health survey[M:48]. Traditional training and assessment typically occurs in optimum daytime working conditions, which arguably[P:44] neglects nocturnal circadian effects. One study[P:44] found that training delivered during a simulated night shift in an emergency department was effective in significantly increasing medical students' self-efficacy. Another study[P:48] used electroencephalographic recordings and measured the

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competence of doctors in the course of 24-hr shifts with and without afternoon rest, demonstrating nocturnal cognitive deactivation without rest, and a stimulating vigilance promoting effect of the resting period. Only 2/10 psychophysiological measures[M:129,130] were IM, both were regulatory assessments.

Figure 2 reports the distribution of domains of competence assessed in selected studies reporting in-practice assessments of competence (% of IM), and those from research studies (% of RM). There was notably some variation between the IM and RM in domain prevalence. There was less focus on the 'knowledge' domain in the RM studies. However, the largest discrepancy between the RM and the IM papers was for the 'professionalism' domain, which was rarely included (n = 9) in RM compared to IM (n = 36) ($\chi^2 = 26.6$; $p < 0.001$). **Figure 3** presents the percentage of the procedures that were used in the measures within each of the twelve domains.

Instrument Properties

Instrument properties were evaluated for their attention to the reporting of measurement properties such as reliability, validity, and interpretability. Despite the preponderance of identified measures, there is a dearth of evidence for their reliability and validity; 53.5% of the identified measures provided reliability and validity data (37/73 IM and 45/80 RM), whereas 21.5% of studies described measures without reference to either their reliability or validity (16/73 IM and 17/80 RM).

Measurement content validity was predominantly based upon citation in several of the measures[e.g.M:46,47,48,50,52,64,65,114] rather than validation against an agreed standard measure. With regard to internal consistency, a Cronbach's alpha (α) value was reported for 39 reviewed measures. Cross-cultural validity was addressed in 29.4% of measures. Predictive validity was addressed by a small number of studies to measure the predictive value of pre-medical school assessment on later performance, and the predictive value of cognitive assessment in predicting subsequent impaired competence in aging doctors[P:81,M:123]. Computerised, written, and oral measures were typically single-domain measures of knowledge but reported higher levels of validity.

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Details on measurement interpretability were found for 119 reviewed measures. As many established measurement methods were institution-specific there was substantial variability in testing procedures, content domains and subsequent outcomes. However, practice and test-retest effects in longitudinal studies received minimal consideration[P:42, 81]. Testing measures of competence in real practice using double-blind observations are evidently unfeasible for logistical and ethical reasons, however workplace-based 360-degree feedback from a doctor or trainees colleagues provides an alternative whereby varied sources of in-practice observations with real patients and there is evidence for the validity of this method[M:33,43].

Measurement Assessment Context

Increasing effort has been dedicated to enhancing the realism of technical skills training by introducing SP's and role-plays[P:51]; 45.4% of the reported measures (53/73 IM and 30/80 RM) were designed to be assessed in a simulated context, and just 20.9% with real patients (only 18 of which were IM), 33.7% of measures used neither. The majority assessed competence in either a workplace-based assessment (n = 19), or (more commonly) a simulated assessment (n = 67). However, the acceptability and feasibility of real patient measures rather than simulated context assessment encounters has significant cost, time and ethical barriers.

Positional Orientations

Studies assessed competence measurement from two positional orientations; assessor and assessee. Measurement of assessee competence was categorised by four populations; medical school applicants (n = 3), medical students (n = 60), junior doctors (n = 19) and fully trained (senior) doctors (n = 20). The assessment of medical school applicants for core competencies received minimal attention (3 studies) in the selected literature; one research study[P:9] utilized a video-based situational judgement test (SJT)[M:16] assessing social competencies, the second[P:58] addressed an assessment framework measure[M:91], and the third[P:60] employed a coping responses inventory[M:94] in addition to a state entrance examination[M:95]. Medical students were the target population in the majority of studies, as a competency-based curriculum is becoming a dominant organizing framework for medical education and assessment[P:47], and its emphasis was reflected in its research dominance in the literature. Only

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three studies examined multiple assessee populations; one[P:60] studied medical school applicants, medical students, junior doctors, and fully qualified doctors using psychometric burnout[M:94] and knowledge test[M:95] measures, and found that the assessment of coping competence at medical school enrolment was a predictor of longitudinal professional competence and career development. The second[P:72] (medical students, junior doctors, and fully qualified doctors) used a group evaluation[M:113] to explore the contextual-environmental factors that affect competence assessment, and the third study[P:92] found that vignettes[M:135], SP's[M:136], and chart abstraction[M:137] measures were a valid and comprehensive method to measure competence in junior doctors and fully qualified doctors.

Assessor perspectives are important in the interpretation of competence outcomes. Assessor-performed analogue rating scales were used in 47 measures (31 IM and 16 RM). Evaluative standpoints of assessors varied between the selected studies and were described as follows: Fifty measures (54.8% of IM, 12.5% of RM) employed supervisory occupational appraisal of a junior or student. Six measures (4.1% of IM, 3.8% of RM) used peer assessment. Twenty-two measures (19.2% of IM, 10% of RM) utilised patients or SP's who can provide a key role in evaluating competence as judged by their direct experience of the doctor or trainee. Four measures (2.7% of IM, 2.5% of RM) used disciplinary/regulatory assessment procedures designed to assess a comprehensive range of domains. Finally, 37 measures (8.2% of IM, 38.8% of RM) involved self-assessment of competence. Therefore, a diverse range of assessor standpoints influence competence assessment and providing feedback is arguably[P:23] a complex affective process and the assessment is determined by the assessor's cognition, beliefs, and emotions. Inter-rater reliability is pertinent to the interpretability of selected measures in this review as the majority of multi-domain IM evaluations of clinical performance, such as OSCE's, and the mini-CEX rely extensively on rater-assessment. Inter-rater reliability was reported for a small number of studies[e.g.P:51,67,75,77,89], however many were monocentric studies.

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DISCUSSION

This systematic review sought to identify and evaluate measures of competence in doctors and medical students. 105 articles were identified that included a broad range of 153 competence assessment measures. The principal findings were as follows: Very few studies conceptualised competence through definition or operationalising the construct. Therefore, although competence is considered paramount for safe practice, no consensus in the selected literature exists regarding its definition, nor is there measurement standardisation. Twelve domains used to conceptualise and measure competence were identified. The deconstruction of each domain and its relevance yielded strong operational purpose in 11 of the domains but identified inconsistencies and procedural subjectivities regarding Professionalism (Domain 5). Assessment of professionalism in the studies lacked reliable measurement evaluation tools and was open to implicit judgment biases; this review therefore questions whether further objective assessment of professionalism may be identified and validated. In contrast, judgement and bias (Domain 3) included a range of specific measures applied to assess competence with diversity and judgement orientations; the findings that patient characteristics were commonly associated with treatment outcomes requires continuous examination from a training and assessment perspective to ensure non-discriminatory ethical treatment of all patients[10].

However, it is important to acknowledge that the competence domain categories are not necessarily indicative or proportionally reflective of all globally accepted assessment tools. There are likely varying conceptualisations of competence in different cultures, therefore further research on cultural construct analysis is advocated to explore potential differences in worldwide assessments.

Interestingly, the distribution of competence domains differed between IM and RM; the RM reflecting more explicit and focused consideration of the contribution of behaviouristic (skills and performance), systemic pressures, psychophysiological (cognitive, emotional, health), and vantage-point (beliefs, perceptions and attitudes) factors as conceptually relevant to competence. A further follow-up exploration to examine which of the RM are being implemented in systemic operational practice would be worthwhile. It is noteworthy that findings related to the validation of the measures were often limited or entirely absent, therefore

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further validation of existing measures is recommended. There is a broadly held assumption that assessment formats with singular competence domains are limited[11] whereas utilising comprehensive multidimensional assessment measures will minimise deficiencies[12]. Furthermore, the majority (n = 76) of selected studies were single-centre assessments. To address this, we encourage multicentric testing of competence assessment to offer greater generalisability of the data. Correspondingly, the sharing of assessment measures between institutions may encourage greater homogeneity in practices.

Assessor bias is a worthwhile area of further research due to concerning discrepancies in assessment accuracy that have been attributed to unreliable self-assessment,[13] inconsistent applications of organizational standards resulting in a lack of reliability, and unclear rating scale criteria. Assessments involving direct observations inherently include automatic judgement processes, and don't necessarily provide objective assessment[14]. However, it has been argued,[15] that standardisation may be a constraint on authentic assessment interpretations and delineations. Approaches using 360-degree global rating evaluations by multidisciplinary multilevel colleagues who interact with a doctor or trainee routinely may improve objectivity through a comprehensive range of perspectives compared to single-assessor perspectives[16]. This review highlighted the inconsistent judgement acuity of assessors as paramount to competence assessment[e.g.P:1,5,7,17,18,20,28,36,47,52,69,101]. When evaluating instrument properties, the consideration of feasibility is paramount, as all measures require resources for systemic implementation. Furthermore, the number of individuals being assessed, the time constraints, financial resources, the interpretability, the validity and reliability of measures are key considerations. When reviewing these points, many experimental measures are unfeasible as routine in-practice measures due to resource availability and/or ethical and time constraints.

There was unanimity in the literature that competence assessment throughout a doctor's career is necessary to protect patients and the profession,[1]. However, this review has not identified a single-best-tool. Therefore, continued exploration of evidence-based measures to assess competence throughout doctors' training and career is advocated. This review highlights the absence of consensus on the domains relevant to competence, and a lack of consistency in existing clinical measurement standards. In conclusion, the findings of this study strongly support the use of multidomain evidence-based measurement-selection, and attending to

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developing measures that replicate the unique systemic context through the evaluation of all domains relevant to competence is recommended as a continued focus of research in this area.

Contributors All authors developed the protocol for this systematic review. KH conducted the screening of studies, data extraction and critical appraisal. DL and AL reviewed each stage of study selection and calibrated findings. All authors assisted in the interpretation and write up of results. All authors approved the final manuscript prior to submission.

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FIGURE LEGENDS

Figure 1 reports the flow diagram of the review search strategy and study selection based on PRISMA (2009) guidelines

Figure 2 reports the distribution of domains of competence assessed in selected studies reporting in-practice assessments of competence, and those from research studies

Figure 3 presents donut charts representing the percentage of the procedures that were used in the measures within each of the twelve domains

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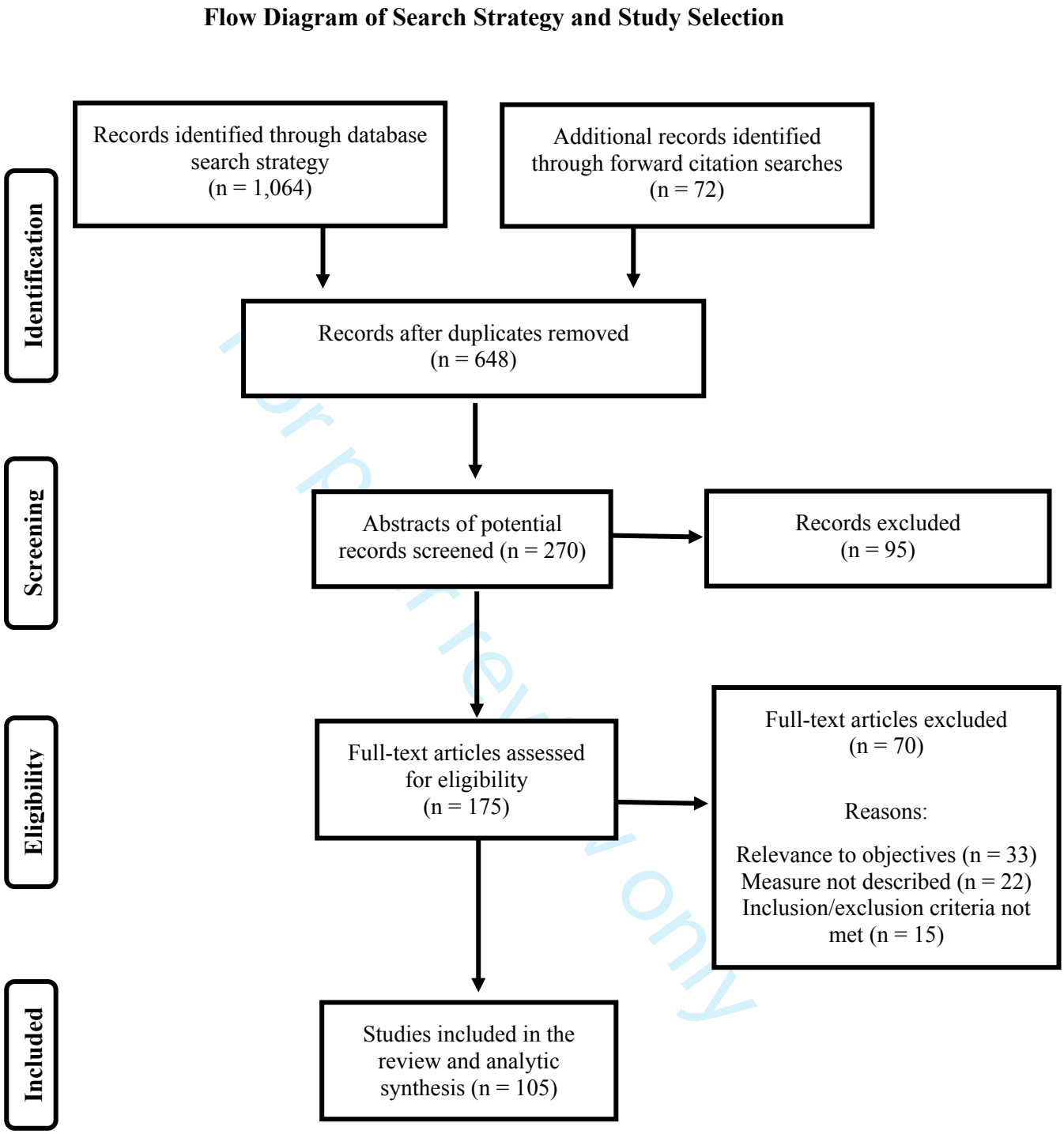


Figure 1 Flow Diagram of Search Strategy and Study Selection Based on PRISMA (2009) Guidelines

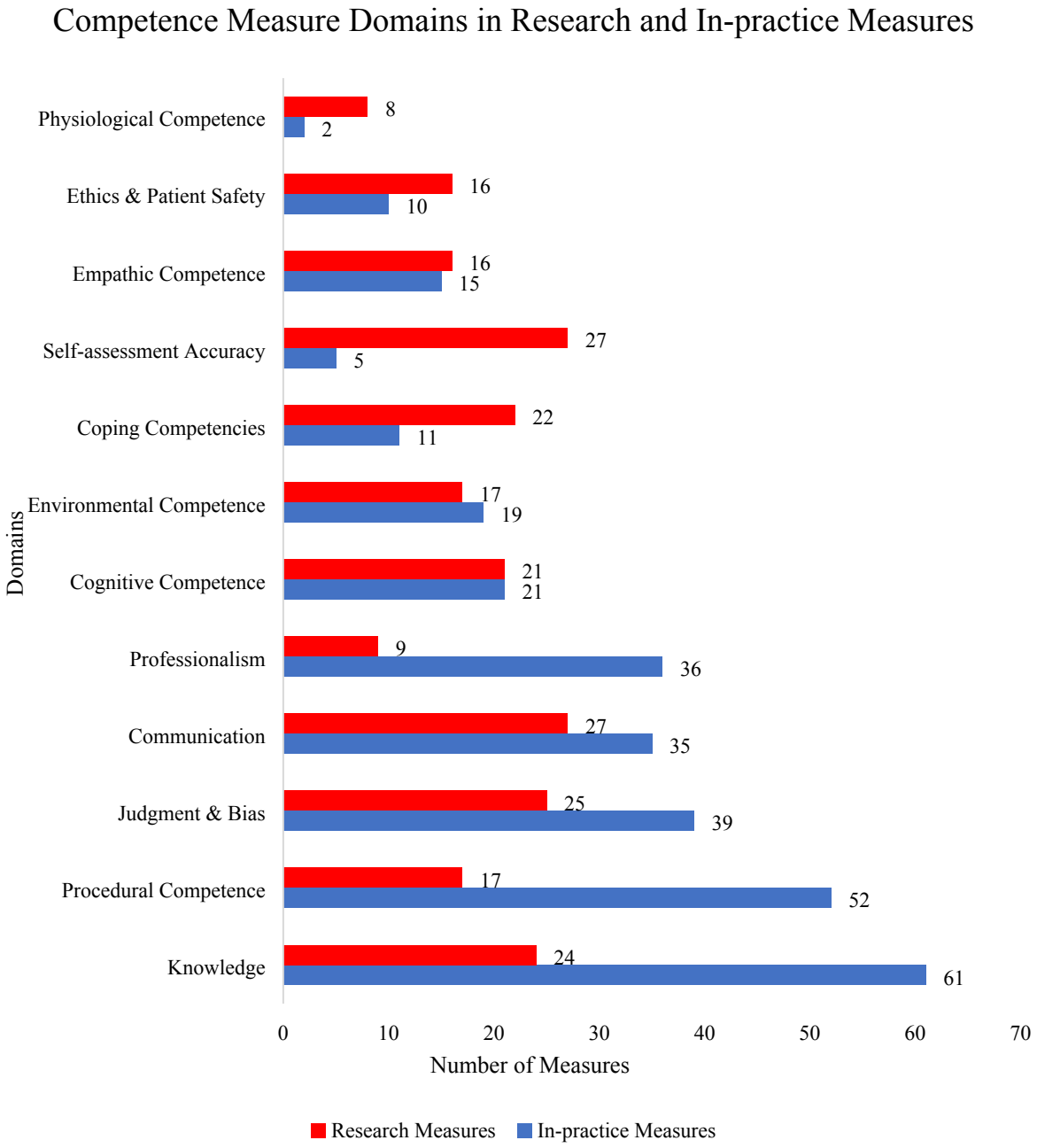


Figure 2 Competence Domains in Research and In-practice Measures

Procedural Variation Within Each of the 12 Domains

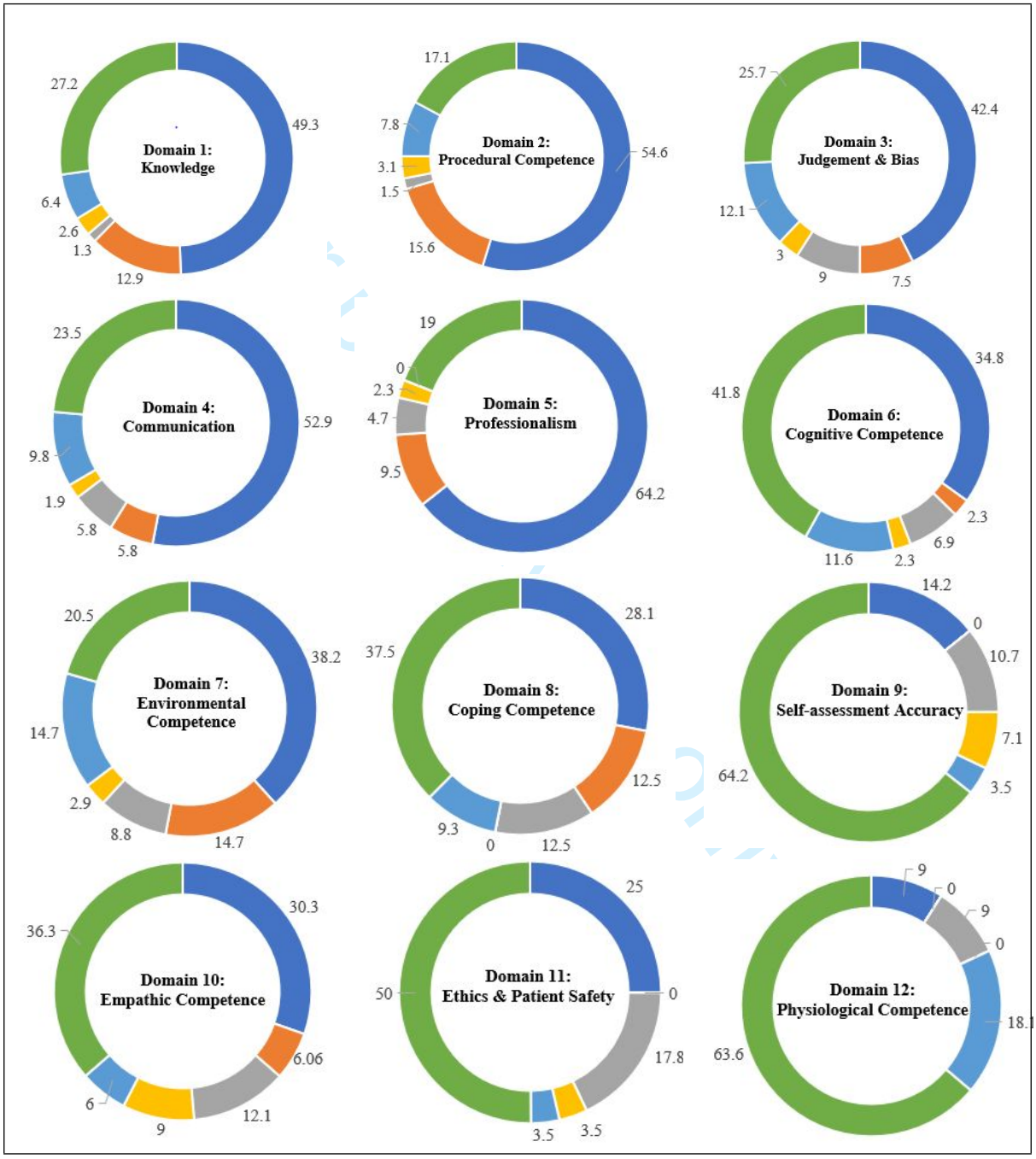


Figure 3 Donut charts representing the distribution of procedural variation within each competence domain

MEASURING COMPETENCE IN DOCTORS

Supplemental Material File 1

Study Search and Selection Results

Electronic Search Engine		Total Identified Records	Number Following Deduplication	Relevant Titles Selected for Abstract Screening	Full Text Articles Assessed for Eligibility	Studies Included in Analytic Synthesis
PsychSource	English Language Search	221	221	31	17	7
	American-English Search	224	42	12	9	6
Medline/ PubMed	English Language Search	132	65	35	19	10
	American-English Search	179	57	39	14	9
Cochrane Library – Central	English Language Search	53	41	20	19	13
	American-English Search	117	56	17	18	11
Web of Science	English Language Search	58	52	19	14	10
	American-English Search	80	42	25	15	11
Specific Paper Retrieval at Journal Site – Forward Citation Search		72	72	72	51	28
Total		1, 136	648	270	175	105

Note. Search results are reported in the sequence they were undertake

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Supplemental Material File 2

Assigned Reference Identifier Codes and Full References

Identifier	Reference	Identifier	Reference
P1	Boursicot (2006)	P46	ALQahtani *et al.*, (2018)
P2	Hojat *et al.*, (2002)	P47	Yeates *et al.*, (2012)
P3	Ogle *et al.*, (2013)	P48	Frey *et al.*, (2002)
P4	Maudsley *et al.*, (2007)	P49	Huizinga *et al.*, (2018)
P5	Wilkinson & Frampton (2003)	P50	Aper *et al.*, (2012)
P6	Williams & Klamen (2015)	P51	Nikendei *et al.*, (2007)
P7	Morgan & Cleave-Hogg (2002)	P52	Hall *et al.*, (2015)
P8	Burgess *et al.*, (2011)	P53	Truncali *et al.*, (2011)
P9	Frölich *et al.*, (2017)	P54	Chumley *et al.*, (2006)
P10	Dahlin *et al.*, (2010)	P55	Ludwig *et al.*, (2018)
P11	Kell *et al.*, (2014)	P56	Windish *et al.*, (2005)
P12	Wilkinson & Frampton (2004)	P57	Bugaj *et al.*, (2018)
P13	Oh *et al.*, (2016)	P58	Schreurs *et al.*, (2018)
P14	Homberg *et al.*, (2019)	P59	Gude *et al.*, (2017)
P15	Monteiro *et al.*, (2019)	P60	Tartas *et al.*, (2016)
P16	Radabaugh *et al.*, (2019)	P61	Han & Chung (2016)
P17	Harendza *et al.*, (2019)	P62	Devine *et al.*, (2015)
P18	Madrazo *et al.*, (2018)	P63	Roland *et al.*, (2015)
P19	Woods *et al.*, (2018)	P64	Carr *et al.*, (2014)
P20	Prediger *et al.*, (2019)	P65	Mehdizadeh *et al.*, (2014)
P21	Gerard *et al.*, (2018)	P66	Ammentorp *et al.*, (2013)
P22	Renaud & Côté (2017)	P67	Terregino *et al.*, (2019)
P23	Sehlbach (2020)	P68	Zelin *et al.*, (2018)
P24	Olsen *et al.*, (2019)	P69	Pool *et al.*, (2018)
P25	Burnett *et al.*, (2019)	P70	Patterson *et al.*, (2017)
P26	Petrosoniak *et al.*, (2019)	P71	Zabar *et al.*, (2016)
P27	Siau *et al.*, (2019)	P72	Stroud *et al.* (2015)
P28	Kain *et al.*, (2019)	P73	Park *et al.*, (2015)
P29	Prediger *et al.*, (2020)	P74	Dong *et al.*, (2015)
P30	Giesler *et al.*, (2011)	P75	Ginsburg *et al.*, (2015)
P31	Bodamer *et al.*, (2015)	P76	Ferenchick & Solomon (2013)
P32	Beckman *et al.*, (2012)	P77	Thomas *et al.*, (2011)
P33	Hautz *et al.*, (2017)	P78	Ginsburg *et al.*, (2010)
P34	Krautter *et al.*, (2011)	P79	Jani *et al.*, (2017)
P35	Liddell *et al.*, (2002)	P80	Spittal *et al.*, (2015)
P36	Yazbeck-Karam *et al.*, (2014)	P81	Turnbull *et al.*, (2006)
P37	Engerer *et al.*, (2016)	P82	Caulford *et al.*, (1994)
P38	Posner *et al.*, (2013)	P83	Grace *et al.*, (2014)
P39	Mc Evoy *et al.*, (2016)	P84	Lehmann *et al.*, (1990)
P40	Snow *et al.*, (2016)	P85	Meagher *et al.*, (2009)
P41	Hoefer *et al.*, (2017)	P86	Mehdizadeh *et al.*, (2014)
P42	Shapiro *et al.*, (2009)	P87	Unwin *et al.*, (2019)
P43	Krautter *et al.*, (2014)	P88	Devitt *et al.*, (1998)
P44	Stroben *et al.*, (2016)	P89	Joshi *et al.*, (2004)
P45	Christiansen *et al.*, (2014)	P90	Smith *et al.*, (2019)

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Identifier	Reference	Identifier	Reference
P91	Santen *et al.*, (2013)	P99	Green *et al.*, (2017)
P92	Peabody *et al.*, (2000)	P100	Mitchell *et al.*, (2011)
P93	Humphrey *et al.*, (2011)	P101	Tham *et al.*, (2017)
P94	Gollehon *et al.*, (2017)	P102	Shulruf *et al.*, (2019)
P95	Austin *et al.*, (2007)	P103	Shi *et al.*, (2018)
P96	Kirby *et al.*, (2011)	P104	Trockel *et al.*, (2018)
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MEASURING COMPETENCE IN DOCTORS

Supplemental Material File 3

Comprehensive List of All Identified Measures

Reference Identifier	IM/ RM	Outcome Measure	Measure Identifier
P1	IM (A)	The Angoff Method (OSCE rating scale)	M1
P2	IM	US Medical Licensing Examinations (USMLE)	M2
P2	IM	Medical College Admission Test (MCAT)	M3
P2	RM	Jefferson Scale of Physician Empathy	M4
P3	RM	Jefferson Scale of Physician Empathy – Student Version (JSPF-S)	M5
P3	IM	Objective structured clinical examination (OSCE)	M6
P3	IM	Rating Scales for the Assessment of Empathic Communication in Medical Interviews (REM)	M7
P4	RM	Questionnaire survey about notions of a 'good doctor' 6 brief questionnaire surveys	M8
P5	IM (A)	Performance Rating Instrument - A Trainee Intern Rating Form	M9
P6	RM	Twenty Questions (TQ) game performance	M10
P6	IM	Medical College Admission Test (MCAT) Biological Science Test Score	M11
P7	RM	25-point prior experience and confidence questionnaire	M12
P7	IM	A pre-determined critical event simulation	M13
P8	RM	Vignettes on chronic pain (assessed by treatment outcome)	M14
P8	IM	The short Medical Authoritarianism Scale (MA-S)	M15
P9	RM	A video-based situational judgment test (SJT) assessing social competencies	M16
P10	RM	Performance-based self-esteem (PBSE)	M17
P10	RM	Oldenburg Burnout Inventory (OLBI)	M18
P11	IM	Standardised Patient Examination (SPE)	M19
P11	IM	Opinions about Physicians Interactions with Patients (OPIP)	M20
P11	IM	Medical College Admission Test (MCAT) GPA	M21
P12	IM	OSCE combined with a written examination	M22
P13	RM	Anderson's Epistemological Beliefs Questionnaire (EBQ)	M23
P13	IM	SP: Korean Clinical Performance Examination (CPX)	M24
P14	IM (A)	The CanMEDS Physician Competency Framework	M25
P15	IM	Written Vignettes	M26
P15	RM	Time-pressured Vignette Assessment	M27
P16	IM	US Medical Licensing Examinations (USMLE)	M28
P17	RM	Group Assessment Performance (GAP) Test	M29
P18	IM	OSCE self-assessment measure on a 6-point Global Rating Scale (GRS)	M30

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P19	RM	Workload InsightTM Application (Chronometry measure of task)	M31
P19	IM	The Secondary Task Method – Venipuncture (Simulated task performance rated using a 16-point checklist)	M32
P20	RM (A)	A 360-degree Assessment of Competencies	M33
P21	RM	PediatricSim Electronic Game	M34
P22	RM	The 5-item Doctor-Patient Communication Competency Instrument for Medical Students (SP)	M35
P23	RM	Patient Evaluation of Physician Competence	M36
P24	RM	Diagnostic Competency Framework	M37
P25	IM	Simulated Patient (SP) with simulated crisis parameters of mortality and morbidity	M38
P26	IM	7-point Global Rating Scale (GRS)	M39
P26	RM	Chronometry	M40
P27	IM	Direct Observation of Procedural Skills (DOPS)	M41
P28	RM (A)	Qualitative Experiential Knowledge	M42
P29	RM	360-degree Assessment	M43
P30	RM	The Freiberg Questionnaire to Assess Competencies in Medicine	M44
P31	RM	Internal Medicine Standardized Simulation Based Examination (SSBE)	M45
P32	RM	The Interpersonal Reactivity Index (IRI)	M46
P32	RM	The Maslach Burnout Inventory	M47
P32	RM	Medical Outcomes Study Short Form (SF-8)	M48
P32	IM	The Mini-Clinical Evaluation Exercise (mini-CEX)	M49
P32	IM	In-training Examinations (ITE)	M50
P33	IM	Global Rating Scale (GRS)	M51
P33	RM	The Test of Self-Conscious Affect, Version 3 (TOSCA-3)	M52
P34	RM	Chronometry	M53
P34	RM	Objective Performance Ratings by Independent Assessors of Videographed Procedure	M54
P35	IM	Procedural Assessment of Clinical Skill (Injection & Suturing)	M55
P35	RM	Procedural Skills Confidence Survey	M56
P36	IM	OSCE (SP)	M57
P37	IM	Student Self-Ratings of Communication Competence (VAS)	M58
P38	IM	SP-PTT (part-task trainer) hybrid: OSCE	M59
P38	IM	PTT (part-task trainer): OSCE	M60
P39	RM	Written reflective assignment	M61
P39	RM	Rating Scale - self assessment	M62
P40	IM	OSCE	M63
P40	IM	Doctors' Interpersonal Skills Questionnaire (DISQ)	M64
P41	IM	OSCE Station (videotaped)	M65
P42	RM	Self-Assessment of Interpersonal Competence Questionnaire (SAICQ)	M66
P42	RM	The Staff-Patient Interaction Scale (SPIR)	M67
P42	RM	Interpersonal Skills Rating Scale (ISRS) rated by SP's	M68

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P43	RM	Self efficacy rating (Likert scale)	M69
P43	IM	SP clinical performance assessment by assessor direct observation (DO)	M70
P44	RM	Self-reported preparedness (self efficacy) Likert questionnaire	M71
P44	IM	SP observation using the Calgary-Cambridge Observation Guide (CCOG)	M72
P45	IM	Radiological (X-ray) diagnostic accuracy	M73
P45	RM	Diagnostic Confidence Rating (Likert scale)	M74
P46	RM	Diagnosis using written clinical vignettes on Qualtrics	M75
P46	RM	Chronometry	M76
P47	IM (A)	mini-CEX	M77
P48	RM	Computer-analyzed Electroencephalography	M78
P48	RM	Pauli Test	M79
P49	RM	The Mini-NeuroCart	M80
P50	RM	Self efficacy scale	M81
P50	RM	Responses to a standardised video case	M82
P51	IM	OSCE test - Global rating scale of communication skills - binary checklist	M83
P52	RM	Analogue Patient Satisfaction Questionnaire	M84
P53	IM	OSCE SP checklist	M85
P54	IM	SP case	M86
P55	IM	Formative exit exam (medical school module) on clinical reasoning and diagnostic accuracy	M87
P56	IM	SP's who rated students on a 5-point scale for communication skills	M88
P56	IM	Diagnostic Thinking Inventory	M89
P57	IM	Procedural Protocol Instrument (IPPI) Global rating scale - blind video assessed	M90
P58	RM	CanMEDS Assessment Framework Measure	M91
P59	RM	The Oslo Inventory of Self-Reported Communication Skills (OSISCS)	M92
P59	IM	SP - The Arizona Communication Interview Rating Scale (ACIR)	M93
P60	RM	Coping Responses Inventory (Moos)	M94
P60	IM	State Examination for Medical Doctors	M95
P60	RM	Maslach Burnout Inventory (MBI)	M96
P61	IM	CPX - Clinical Performance Examination (SP)	M97
P61	IM	OSCE	M98
P62	IM	MRCP (UK)	M99
P62	IM	MRCGP AKT (Applied Knowledge Test)	M100
P63	RM	Self-evaluation meta-planning focus group	M101
P64	IM	The Junior Doctor Assessment Tool (JDAT)	M102
P65	IM	GMC test of Competence (ToC)	M103
P65	IM	OSCE	M104
P65	RM	Self-Assessment Questionnaire - ranking in comparison to other doctors	M105
P66	IM	OSCE - using the Calgary-Cambridge Observation Guide Checklist (SP and observer)	M106

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P66	RM	Self-assessment - Likert scale	M107
P67	RM	Empathy and Clarity Rating Scale (ECRS) in OSCE evaluation	M108
P68	RM	Comfort and competence of working with sexual and gender minorities survey	M109
P69	IM (A)	Portfolio-based assessment	M110
P70	RM	Wechsler Adult Intelligence Scale (WAIS-IV)	M111
P71	RM	OSCE - Standardized nurse (SN) rating on professional collaboration	M112
P72	RM	Focus groups	M113
P73	IM	OSCE - 4 stations (SP)	M114
P73	RM	5-minute examination - Diagnostic Accuracy	M115
P74	RM	The Essential Elements of Communication (EEC) - rating scale on an OSCE (SP)	M116
P75	RM	Socio-cultural domains of patient safety - OSCE (SP)	M117
P76	RM	Electronic 'Just in Time Medicine' (JIT) clinical evaluation exercize (CEX)	M118
P77	IM	Group Assessment using faculty opinion survey	M119
P78	IM (A)	Assessor Rating Scale	M120
P79	IM	360-degree feedback	M121
P80	RM	PRONE score	M122
P81	RM	PREP assessment (Physician Review Program) Test Battery	M123
P82	RM	Chart stimulated recall (CSR)	M124
P82	IM	SP's who rated students on a 5-point scale for communication skills	M125
P83	IM	Center for Personalized Education for Physicians (CPEP) multivariate competence assessment	M126
P84	RM	Patient-physician Interaction Quality Checklist	M127
P85	RM	Team Objective Structured Bedside Assessment (TOSBA)	M128
P86	IM	GMC Test of Competence (ToC) - including a written knowledge test and an OSCE	M129
P87	IM	GMC Test of Competence (ToC) - including a written knowledge test and an OSCE	M130
P88	RM	Knowledge-based questionnaire	M131
P89	RM	360-degree Assessment	M132
P90	IM	OSCE (SP)	M133
P91	RM	Emergency Medicine Milestones Survey	M134
P92	IM	Vignettes	M135
P92	IM	SP's	M136
P92	IM	Chart abstraction	M137
P93	IM	Referral for fitness to practice investigation	M138
P94	IM	OSCE	M139
P95	RM	The Jefferson Scale of Physician Empathy (JSE)	M140
P96	RM	The Scale of Attitudes Toward Disabled Persons (SADP)	M141
P97	RM	The Jefferson Scale of Physician Empathy (JSE)	M142
P98	RM	The Almost Perfect Scale-Revised (APS-R)	M143
P98	IM	Grade Point Average (GPA)	M144

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P99	RM	The cross-cultural care survey (CCCS)	M145
P100	IM	Workplace Based Assessments (WPBA) including: mini-clinical evaluation exercize (mini-CEX), case-based discussion (CbD), direct observation of procedural skills (DOPS), and mini-peer assessment tool (mini-PAT)	M146
P101	IM (A)	Workplace Based Assessments (WBA's)	M147
P102	IM	OSCE	M148
P103	RM	Support vector machine combined with an ordinal partitioning model (SVMOP)	M149
P104	RM	Professional Fulfilment Index (PFI)	M150
P105	RM	Epworth Sleepiness Scale	M151
P105	RM	QOL Burnout and depression	M152
P105	RM	Self-reported medical errors	M153

Notes.

EM: Existing Measures, RM: Research Methods, (A): Assessor assessment.

Papers in the manuscript are referenced in the format $^{P1,2,3,4,5\ etc.}$ individual measures are referred to as ^{M1,2,3,4,5} etc.

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Supplemental Material File 4

List of Measures in Each Domain

Domain 1				Domain 2				Domain 3			
M1	M33	M72	M117	M2	M41	M88	M129	M2	M51	M100	M147
M2	M34	M73	M118	M6	M43	M90	M130	M4	M52	M102	
M6	M36	M75	M119	M8	M45	M93	M134	M5	M57	M103	
M8	M37	M82	M124	M9	M49	M95	M136	M6	M59	M104	
M9	M38	M83	M125	M13	M50	M97	M139	M8	M60	M106	
M10	M39	M85	M126	M14	M51	M98	M148	M9	M63	M109	
M11	M41	M86	M128	M19	M54	M99		M10	M65	M111	
M12	M43	M87	M129	M20	M55	M100		M13	M70	M114	
M13	M44	M88	M130	M22	M56	M102		M14	M72	M115	
M14	M45	M89	M131	M24	M57	M103		M15	M73	M118	
M17	M49	M90	M133	M25	M59	M104		M19	M74	M119	
M19	M50	M95	M134	M28	M60	M106		M28	M75	M126	
M21	M51	M97	M135	M29	M63	M114		M29	M77	M129	
M22	M54	M98	M136	M30	M65	M115		M31	M84	M130	
M24	M55	M99	M137	M32	M70	M117		M33	M87	M135	
M25	M57	M100	M139	M33	M72	M119		M34	M88	M136	
M26	M59	M102	M144	M34	M73	M122		M36	M90	M139	
M27	M60	M106	M146	M36	M77	M124		M37	M91	M141	
M28	M63	M109	M147	M37	M83	M125		M38	M97	M143	
M30	M65	M114	M148	M38	M85	M126		M43	M98	M145	
M32	M70	M115		M39	M86	M128		M45	M99	M146	

Domain 4			Domain 5			Domain 6		Domain 7		Domain 8	
M8	M67	M112	M1	M98	M139	M3	M75	M8	M75	M18	M94
M9	M68	M116	M22	M99	M148	M8	M79	M13	M76	M27	M96
M16	M83	M117	M33	M100	M150	M9	M80	M18	M90	M29	M103
M20	M84	M118	M36	M102		M10	M82	M19	M91	M31	M104
M25	M85	M119	M38	M103		M11	M89	M25	M103	M32	M113
M29	M86	M121	M41	M104		M21	M91	M27	M104	M33	M129
M33	M88	M125	M43	M106		M22	M99	M29	M113	M38	M130
M35	M92	M126	M44	M114		M23	M100	M31	M117	M43	M134
M36	M93	M127	M51	M115		M24	M103	M32	M119	M44	M139
M37	M95	M128	M52	M117		M26	M104	M33	M126	M47	M143
M38	M97	M129	M59	M119		M27	M111	M37	M129	M48	M150
M41	M98	M130	M60	M121		M29	M117	M38	M130	M51	M152
M43	M99	M132	M63	M125		M31	M118	M40	M133	M52	
M44	M100	M133	M65	M126		M32	M123	M43	M136	M56	
M46	M102	M136	M70	M128		M33	M128	M47	M139	M59	
M52	M103	M139	M72	M129		M34	M129	M51		M60	
M58	M104	M146	M83	M130		M37	M130	M52		M71	
M59	M106	M147	M85	M133		M38	M139	M53		M75	
M60	M108	M148	M90	M136		M41	M144	M59		M80	
M64	M109	M149	M95	M137		M43	M146	M60		M90	
M66	M111		M97	M138		M44	M147	M72		M91	

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Domain 9		Domain 10		Domain 11		Domain 12
M8	M96	M4	M121	M1	M138	M8
M12	M101	M5	M126	M4	M146	M18
M17	M105	M7	M129	M5	M147	M31
M18	M107	M8	M130	M8	M150	M48
M23	M113	M9	M132	M9	M153	M78
M25	M124	M14	M136	M14		M80
M30	M126	M15	M140	M15		M129
M33	M129	M36	M142	M23		M130
M43	M130	M43	M146	M36		M150
M52	M134	M46	M147	M43		M151
M56	M143	M58		M52		
M61		M59		M61		
M62		M60		M62		
M66		M61		M91		
M69		M62		M117		
M71		M64		M121		
M74		M84		M122		
M80		M91		M123		
M81		M106		M126		
M92		M108		M129		
M94		M109		M130		

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6-7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7-8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8-9

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Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8-9
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-15
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10-19
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	20-22
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	21
FUNDING			

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Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22

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The variable trajectory: A systematic review, analytic synthesis and construct domain consolidation of international measures of competence in doctors and medical students.

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MEASURING COMPETENCE IN DOCTORS

THE VARIABLE TRAJECTORY: A SYSTEMATIC REVIEW, ANALYTIC SYNTHESIS AND CONSTRUCT DOMAIN CONSOLIDATION OF INTERNATIONAL MEASURES OF COMPETENCE IN DOCTORS AND MEDICAL STUDENTS

Running title: Measuring competence in doctors

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The variable trajectory:
A systematic review, analytic synthesis and construct domain consolidation of international measures of competence in doctors and medical students

ABSTRACT

Background

Competence is assessed throughout a doctor's career. Failure to identify and manage impaired competence can have critical consequences. Consistent conceptualisation and accurate measurement of this construct is imperative. Therefore, the objective of this review was to identify and evaluate measures used to assess competence in doctors and medical students.

Methods

A systematic search of the published literature was undertaken between December 2019 and February 2020 for articles reporting on the measurement of competence in doctors and/or medical students. Searches were conducted in the PsychSOURCE, US National Library of Medicine National Institutes of Health, MEDLINE (PubMed), The Cochrane Central Register of Controlled Trials (CENTRAL), and Web of Science electronic databases. Citation screening and forward citation tracking of included studies was carried out to identify any further relevant papers for inclusion. One thousand one-hundred and thirty-six potentially relevant articles were screened. An analytic synthesis approach was implemented to the identification, organization and interpretation of homogenous study and measure characteristics.

Results

Twelve competence domains were identified from the 153 identified measures. Knowledge and Procedural competence domains were the dominant focus of publications reporting current medical practice, but less so in research-based studies which more frequently assessed interpersonal, psychological, physiological and ethical competencies. In the 105 included articles,

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the reporting of measurement instrument quality was varied, with comprehensive reporting only present in 53.6% of measures; validation for some of the measures was particularly limited.

Discussion

Whilst this review included a considerable number of publications reporting the measurement of competence in doctors and medical students, the heterogeneity of the measures and variation of findings limit the ability to evaluate their validity and generalisability. However, this review presents a resource for researchers and medical educators which may inform operational practice and future research.

Keywords: Competence · Doctor · Medical student · Physician · Performance

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Strengths and limitations of this review

- To our knowledge, this is the first contemporary systematic review to consolidate globally recognised domains relevant to competence assessment of doctors and medical students in both research and in-practice measures.

- This PROSPERO registered review employed an extensive and rigorous search strategy of four databases and forward citation searches using defined selection and abstraction criteria calibrated by three researchers, undertaken in alignment with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement, and guided by the synthesis without meta-analysis (SWiM) reporting guidelines.

- The review identified a broad range of measures used to assess competence, and identifies the homogeneity and heterogeneities using a textual and conceptual analytic synthesis approach.

- A limitation of this review is that English language restrictions were placed on the searches, and although research studies were derived from 18 countries, the inclusion of further languages may yield unidentified studies, and subsequent differences in worldwide assessments of competence.

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INTRODUCTION

Competence is assessed throughout doctors' professional lifespan, from medical student to junior doctor, to senior doctor until retirement. The timely identification, support and remediation of doctors who do not demonstrate sufficient competence to undertake their duties safely is imperative. However, the proportion of doctors who are incompetent, or dyscompetent, and failing to uphold adequate practice standards is unclear[1]. Robust assessment of competence necessitates accurate and standardised measures that comprehensively examine all domains relevant to the core competencies required to undertake a job effectively. It has been argued that competencies are learned, complex context-specific dispositions, but the complexity and diversity inherent in doctors' responsibilities make measurement of competence a complex task[2].

Despite numerous definitions,[3] an agreed explicit, amalgamated, and quantifiable definition of the construct of competence, and in many cases a consistent and transparent chain of inferences to validate the observed assessment and measurement of competence are lacking[4]. It has been argued,[5] that it is challenging to identify reliable and valid measures of competence as many of the skills are considered difficult to quantify. Contemporary competence-based medical education is predominantly driven by patient safety, efficiency and clinical outcomes[6].

In order to explore these challenges, and provide an overview of the diverse facets of competence, and their dominance within current assessment measures; the objective of this systematic review of peer-reviewed scientific literature was to identify and evaluate specific measures of competence relevant to doctors and medical students. As the accurate measurement of competence is paramount to patient safety, a systematic review of the measures employed and domains of competence assessed in medical students and doctors was considered important.

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METHOD

The review protocol was registered with PROSPERO, adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement,[7] and guided by the 'synthesis without meta-analysis (SWiM) reporting guidelines'[8].

Literature Search

Data Sources

A systematic search of the literature was conducted in the PsychSOURCE, US National Library of Medicine National Institutes of Health, MEDLINE (PubMed), The Cochrane Central Register of Controlled Trials (CENTRAL), and Web of Science electronic journal databases. Citation screening and forward citation tracking of included studies was carried out to identify any further relevant papers; these were retrieved from the journal site or academic library databases.

Search Strategy

A pilot search was undertaken to determine relevant electronic search engines, refine the search strategy, define search keywords, identify relevant MeSH terms, determine search restrictions, and select Boolean operators. English and American-English language terms were used. Search terms employed the keywords; 'doctor', 'competence', and 'performance', incorporating the Boolean operator ('and'), in addition to MeSH terms (physicians, medical students). Restrictions on the databases included human studies, published in English language, and full text availability. No date restrictions were applied. Searches were undertaken between December 2019 - February 2020. Further searches in response to the literature were performed through reference lists of reviewed articles in forward citation searches. Results were downloaded to reference management software and de-duplicated.

Study Identification and Selection

Inclusion/Exclusion Criteria

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Studies were included if they described the measurement of competence in medical students and doctors, with full-text availability in peer-reviewed journals, published in English language. Commentaries, editorials, letters, book chapters, conference papers, books, doctoral theses, dissertations, and articles which did not adequately report competence assessment measures were excluded.

Calibration of Findings

One reviewer undertook title and abstract screening, and selection methods were supervised by two reviewers who verified the extraction of data, checked consistency and methodological clarity. Study selection disagreements and subsequent queries about inclusion following full-text review were resolved by discussion between researchers to reach consensus.

Study Selection

From searches, 1,136 papers were identified; after duplicates were removed 648 original articles remained. Following assessment for relevance to the objectives and selection criteria 105 articles were selected for systematic analysis. **Figure 1** reports the PRISMA (2009) flow diagram of study selection, and specific search engine results are presented in *supplementary file 1*.

Data Extraction

Selected articles were subjected to three preliminary levels of screening: Title, Abstract and Full Text. The following data (when available) from articles was extracted using a pre-designed and piloted data-extraction pro-forma which included the following categories and sub-categories: (1) Referencing information, (2) Quality assessment, (3) Study characteristics (geographical origin, sample, objectives, design), (4) Type of measure, (5) Assessment to examine whether the properties of each measurement tool (construct validity, content validity, criterion validity, internal consistency, test-retest reliability, inter-rater reliability etc.) had been reported, (6) Main results/outcomes, (7) Additional further information relevant to the objectives. Results were recorded in an Excel spreadsheet and each study was assigned a reference identifier code (*supplementary file 2*).

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Study Assessment of Methodological Quality

Quality inclusion criteria were restricted by whether sufficient detail was provided to enable analysis of methodological quality and the evaluation of study measures. The five 'appraisal prompts for informing judgements about quality of papers' defined by Critical Interpretive Synthesis (CIS)[9] methodology were evaluated and tabulated for all included studies. CIS yielded quality appraisal nominal data scores for each question (1 yes, 0 no), with a cumulative score ranging from 0 to 5; scores ranged from 2 to 5. Ninety-four of the 105 studies achieved the maximum (5/5) cumulative rating score, six scoring (4/5), four scoring (3/5), and one scoring (2/5).

Data evaluation and synthesis

As the protocol anticipated, due to the heterogeneity of measurement tools in this review, and inconsistencies in methodology and reporting of results, a meta-analysis was not possible. Preliminary categorisation of the studies included data extraction of study characteristics, types of competence measures and measurement properties. A conceptual analytic synthesis of identified dimensions of similarity and difference in defining the construct of competence in this population was undertaken.

Patient and Public Involvement

Patients or the public were not directly involved in the design or planning of this study.

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RESULTS

The search strategy generated 105 articles selected for full data abstraction: 13 from PsychSOURCE, 19 MEDLINE/PubMed, 24 Cochrane Library – CENTRAL, 21 Web of Science, and 28 from specific paper retrieval forward citation searches at the journal site. Selected reviewed papers in the results section are referenced in the format[P:1,2,3,4,5], and individual measures are referred to as[M:1,2,3,4,5] (see *supplementary file 2* for assigned reference identifier codes and corresponding references).

Characteristics of Selected Studies

Included articles were published between 1990 and 2020. There was a progressive increase in the number of publications per year over this period (mean annual publications per year 0.3 from 1990 to 1999, 1.9 from 2000 to 2009, and 8.2 from 2010 to 2019), consistent with increasing research interest and societal awareness of the extent and consequences of medical error[P105]. The study designs included: two intervention, 22 measurement development, 26 measurement evaluation, 17 curriculum development, 40 clinical assessment, 21 training evaluation, and 11 experiential assessment; 26 studies comprised multiple categories.

Studies were restricted to English language, but without restriction on geographic origin and were derived from 18 countries. One study[P26] undertook data collection in two countries (USA and Canada), therefore data was counted in both. The majority of research was from the USA (n = 34 articles), UK (n = 17), Germany (n = 15), and Canada (n = 15). There were five studies from Australia[P3,35,64,80,102], four from the Netherlands[P23,49,58,69], and three from Korea[P13,61,73]. Two studies were from New Zealand[P5,12] and Denmark[P45,66]. Sweeden[P10], Lebanon[P36], Saudi Arabia[P46], Austria[P48], Belgium[P50], Norway[P59], Poland[P60], India[P79], and China[P103] were each represented by a single study. The percentage distribution of all articles geographic origin is presented in **figure 2**.

Competence Assessment Measures

From the 105 articles, 153 measures of competence were identified, 98 of which were unique to that study (a comprehensive list of all measures is presented in *supplementary data file*

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3). The competence assessment measures were categorised as either established 'in-practice measures' (IM) currently implemented for the assessment of doctors and medical students, or 'research-measures' (RM) developed or applied to assess competence for research evaluation, but not at the time of publication employed for occupational competence assessment. Thirty-eight studies examined IM, 45 examined RM, and 22 explored both RM and IM. Most (n = 76) of the selected studies were single-institution assessments - limiting the generalisability of findings, 25 were multicentric (based at multiple institutions), and four were not institution-based.

Measures were categorised by their procedural comparability, and seven dominant operational characteristics were identified: (1) Clinical Performance Assessment - was applied in 30.7% of measures (n = 38 IM, n = 9 RM), typically assessed using simulated patient ratings to assess clinical competencies such as communication skills and diagnostic reasoning in response to presented symptoms; (2) Anatomic Model Simulation – providing simulations of human anatomy in order to test knowledge and used in 6.5% of measures (n = 9 IM, n = 1 RM); (3) Psychometric Assessment – predominantly evaluated personality, aptitude, reasoning and emotionality, and conducted in 7.8% of measures (n = 2 IM, n = 10 RM); (4) Patient Records - to assess the medical record keeping accuracy and completeness, were utilised in 1.3% (n = 2 IM); (5) Computerized Testing – included a variety of computerised assessments, used in 9.8% of measures (n = 1 IM, n = 14 RM); and (6) Written Testing – included multiple-choice and short-answer questions in traditional pen-and-paper testing in 31.4% of measures (n = 19 IM, n = 29 RM). The prevalence of each measurement category demonstrated a preference for the application of written testing and clinical performance assessment in the assessment of competence. Fifty-six (36.6%) measures applied a unidimensional procedure to assess the target domains, and 97 (63.4%) operated multidimensional procedures.

Established In-practice Measures

Competence was assessed through a variety of IM methods. Objective structured clinical examinations (OSCEs) typically involve patients or simulated patients and the most duplicated measure (in 20 articles). OSCEs are frequently considered[P:53] the 'gold standard' for assessing competence skills, as they are high-fidelity, can be reliable, and are regarded as valid because of

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their ability to differentiate between occupational levels of aptitude and specialisms. The use of standardised patients (SPs) endeavours to replicate doctor-patient consultations can identify variation in clinical practice between doctors in a realistic simulated setting. SP single-case assessments were used to measure specific competencies in nine of the studies.

Anatomic model simulation testing, employed in 10 measures, is widely used as a safe method to develop procedural competence in early medical education. However, whether the anatomic-model patients are the most valid means of assessing procedural competence is widely debated[P38]. Workplace-based assessment (WBA) such as mini-clinical evaluations (Mini-CEX) and direct observation of procedural skills (DOPS) are heavily integrated in medical education. However, the Mini-CEX and DOPS lack standardisation, cannot be used with large numbers of participants simultaneously, are costly, time consuming, and transferability of skills is not always guaranteed in a simulated assessment setting; particularly in a workplace with high levels of social, psychological and physiological stress[P:20]. Four studies[P:20,29,79,89] examined 360-degree assessments (i.e. rating evaluations from colleagues within the workplace)

Written tests provide reliable assessment of several domains, particularly knowledge and psychometric assessment. Three studies[P:2,95,97] used the Jefferson Scale of Physician Empathy (JSE), one [P:3] employed the Jefferson Scale of Physician Empathy – Student Version (JSPF-S). Written tests measuring knowledge included two studies using the US Medical Licensing Examinations (USMLE), and three studies that included the Medical College Assessment Test (MCAT) - an assessment used by many medical schools as part of the entry-selection process. However, written tests are considered insufficient to measure behavioural clinical competence [P:20]; therefore, there has been a traditional reliance on the judgement of competence by educators or seniors in demonstration-observation tests of clinical performance skills. The combination of a knowledge test (written assessment) with an applied test of clinical skill (OSCE) was shown to increase predictive validity versus either alone[P:12]. Vignettes (five studies) and written case simulations have been widely used to assess competence in both clinical training and experimental research, because of their ease of administration, negligible cost, and objective quantifiable results. However, they do not assess important social interaction skills such as history taking, nor assess

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competence in a real-world situation including occupational stressors. Amongst IM measures there was significant variability in testing procedures, content domains and inconsistent assessment outcomes, therefore these were not grouped for evaluation purposes.

Contemporary Experimental Research Measures

Electronic-technology has an increasing role in healthcare, but raises new challenges for competence assessment; one study presented an electronic platform[M:149] that automatically assesses a doctor's competence from online textual consultations with patients utilizing a novel machine that provides auto-evaluation based on patient satisfaction following online consultation. In addition to patient evaluations, innovative simulation-based medical education (SBME) and assessment provides alternatives to practice and assessment without risk to patients[P:26].

In an era where doctors' retirement age has extended, further cognitive competence considerations have emerged. Forty-two measures in 37 of the studies assessed cognitive competence, publication dates indicating that research interest in this area has markedly risen in recent years (7.6% of the studies between 2000 and 2009, compared to 26.7% from 2010 to 2020). Neuropsychological screening found[P:81] that many of the doctors who fell significantly below expected competence assessment levels had cognitive impairment sufficient to explain their diminished competence. Another study developed 'the Mini-NeuroCart'[M:80], a cognitive and psychomotor test battery for assessment of subjective and objective measures of alertness, mood, concentration and self-assessed procedural competence, and showed that the competence of postcall surgeons was similar to, or worse than that of ethanol-intoxicated surgeons.

Competence Domains

Studies were categorised according to the competence assessment domains. The list of measures in each domain is presented in *supplemental material file 4*. Twelve ascendant domain categories based upon their prevalence in the selected measures were identified, with descriptors of these in **table 1**.

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Table 1 Domain Descriptors in Descendent Order of Prevalence in All Measures

Number	Domain	Prevalence	Descriptor
1	KNOWLEDGE	55.5%	The acquisition and retention of learning.
2	PROCEDURAL COMPETENCE	45.1%	The ability to perform specific technical skills/procedures required for the occupational obligations.
3	JUDGEMENT & BIAS	41.8%	An ability to nullify or negate judgement biases to competently work in the benefit of patients and colleagues.
4	COMMUNICATION	40.5%	The ability to engage patients and colleagues in effective interaction.
5	PROFESSIONALISM	29.4%	To uphold the expected traditional occupational behavioural standards of the profession, and regulatory code of conduct.
6	COGNITIVE COMPETENCE	27.4%	To effectively utilize the cognitive resources required for the selection and application of skills.
7	ENVIRONMENTAL COMPETENCE	23.5%	This domain relates to the ability to maintain acuity in environmental adversity.
8	COPING COMPETENCIES	21.5%	Competence to psychologically manage the complex occupational exigencies.
9	SELF-ASSESSMENT ACCURACY	20.9%	Self-assessment and accurate self efficacy in continuous self-monitoring of competence.
10	EMPATHIC COMPETENCE	20.2%	Empathic Competence describes the awareness and responsiveness to patients' and colleagues' feelings and experiences.
11	ETHICS & PATIENT SAFETY	16.9%	A commitment to exemplar medical practice by prioritising patient safety and ethical standards.
12	PHYSIOLOGICAL COMPETENCE	6.5%	Physiological competence describes a state of optimal functioning that supports the physical and psychological workplace demands.

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Consideration of Each Domain Within the Literature

Knowledge (**Domain 1**), deemed as the primary proximal determinant of competence[P:11] was the most commonly assessed competency in IM, historically assessed through written examinations[P:31], contemporary measures endeavour to combine knowledge with procedural assessment. Concordantly, only 27.2% of measures conformed to traditional written-test approaches, though 6.4% of computerised test measures were a comparable format.

Within **Domain 2** (Procedural Competence), 54.6% of the measures employed a clinical performance test; other measures included patient vignettes, written or computerised tests (30.2%), and anatomic model simulation (15.6%).

Domain 3 (Judgement and Bias) was addressed in 41.8% of measures and incorporated competence with diversity, and awareness of bias in an occupational context. Cultural competence was found to be lacking in one study[P:8] as a result of an interaction between race and treatment outcomes as there was a higher likelihood of doctors escalating the use of opioids for African American patients. Sexual and gender minority individuals experience high rates of discrimination when seeking healthcare, contributing to patient care disparities[P:68]. Furthermore, doctors reported discomfort when treating people who have disabilities[P:96], as assessed by the Scale of Attitudes Toward Disabled Persons (SADP)[M141].

Domain 4 (Communication): Competent doctor-patient and inter-professional communication is included within this domain. Ineffective communication with colleagues has been associated with reduced patient safety and risk to care quality[P:71], and the most frequent cause of complaints against doctors is related to poor communication[P:42].

Though doctors are expected to uphold the customary standards of professionalism (**Domain 5**), there was no consensus on its definition[P:77], and an absence of explicit standardised factors measured within this domain[P:78]. Most (73.8%) of the measures used subjective rating scales; many within OSCE and WBA. Remarkably, although 36 IM measured professionalism, only 9 RM assessed this domain.

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Cognitive competence (**Domain 6**) denotes the specific cognitive resources required for the selection and application of skills such as diagnostic and clinical reasoning. Wechsler Adult Intelligence Scale (WAIS-IV) assessments have found cognitive resource differences between doctors who experience difficulties, and those who do not[P:70]. The cognitive concept of 'mental workload' perceives the brain having a limited capacity to process stimuli, research has found that high levels of workload are associated with error and poor competence[P:19]. Traditional measurement of task performance does not necessarily predict competence when a combination of tasks must be undertaken under time pressure resulting in cognitive overload, and the 'task-shedding' of key competencies resulting in risk of impaired competence.

Environmental Competence (**Domain 7**) employs measures that account for (or simulate) the context of occupational pressures in assimilating and responding to environmental context[P:72] using real-time information management[P:54]. One study[P:19] examined the effect of multitasking on competence in medical students using a secondary task method employing a venepuncture procedure[M:32] alongside an electronic application[M:31] which demanded a tapping response when the device vibrated, mental workload was measured in time delay and effects were observed on communication deterioration. Measures that included time (chronometry)[P:15,19,26,34,46] highlighted that timed tests increased the reliability and validity of measuring diagnostic accuracy[P:15], as time-pressure has been found to disrupt the dynamic interaction between reasoning systems influencing diagnostic tasks[P46]. Conclusions about competence may be flawed unless contextual environmental factors are considered[P:72].

Doctors are expected to handle limited resources, pressure, and life or death performance outcomes that often carry an emotional load; 'Coping Competence' (**Domain 8**) is therefore paramount[P:72]. Distress and inability to cope has been associated with medical errors[P:105]. Therefore, the ability to cope with environmental, cognitive, physical and emotional workplace demands is key to competence and occupational fulfillment[P:104].

The ability to accurately 'self-assess' and work at a level of occupational aptitude. (**Domain 9**) is key to the identification of difficulties, and safe practice. RM focussed more heavily on self report measures (n = 27) than IM (n = 5). Concerningly, one study[P:3] found that self-rated empathy and observer ratings were not associated, and another[P:7] found no

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correlation between confidence scores and clinical or knowledge abilities. A doctor's ability at any career stage to accurately self-assess their competence acquisition provides assurance of working within scope of practice[P:14].

Medical educators and professional regulators recognise the importance of Empathic Competence (**Domain 10**); strong associations were found between observer ratings of empathy and OSCE scores[P:3]. The prominence of empathic competence within the selected measures was comparable in RM (n = 16) and IM (n = 15). In medical education research[P:95], empathic competence has been demonstrated to increase in male trainees and decrease in females during the course of training. Importantly, empathic competence has been found to be strongly associated with patient satisfaction[P:97].

Domain 11 relates to adherence to ethical standards and commitment to patient safety. This domain was addressed in regulatory assessments[e.g.M:129,130,138], rating scales assessed by senior doctors[e.g.M:1,9,91,117,126], WBA's[e.g.M:146,147], patient evaluations[M:36], and values-based measures such as a questionnaire survey about notions of a 'good doctor'[M:8]. Neuropsychological testing[M:123] has been found to be a strong predictor of risk to competence[P:86]; a consistent decline in competence has also been identified in recertification exams in America related to years since graduation[P:82]. Another study[P:75] developed and tested an OSCE designed to directly assess socio-cultural dimensions of patient safety competency[M:117], demonstrating sufficiently reliable station scores in this domain. Engagement with continuous competence improvement and collaboration with colleagues (rather than isolated practice) has been identified as a protective factor[P:28].

Physiological competence (**Domain 12**) emerged as a domain rarely considered in assessment. RM measures within this domain included psychometric measures of burnout[M:18,52], the Epworth Sleepiness Scale[M:151], and the Medical Outcomes Study Short Form (SF8) psychological and physical health survey[M:48]. Traditional training and assessment typically occurs in optimum daytime working conditions, which arguably[P:44] neglects nocturnal circadian effects. One study[P:44] found that training delivered during a simulated night shift in an emergency department was effective in significantly increasing medical students' self-efficacy. Another study[P:48] used electroencephalographic recordings and measured the

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competence of doctors in the course of 24-hr shifts with and without afternoon rest, demonstrating nocturnal cognitive deactivation without rest, and a stimulating vigilance promoting effect of the resting period. Only 2/10 psychophysiological measures[M:129,130] were IM, both were regulatory assessments.

Figure 3 reports the distribution of domains of competence assessed in selected studies reporting in-practice assessments of competence (% of IM), and those from research studies (% of RM). There was notably some variation between the IM and RM in domain prevalence. There was less focus on the 'knowledge' domain in the RM studies. However, the largest discrepancy between the RM and the IM papers was for the 'professionalism' domain, which was rarely included (n = 9) in RM compared to IM (n = 36) ($\chi^2 = 26.6$; $p < 0.001$).

Figure 4 presents the percentage of the procedures that were used in the measures within each of the twelve domains. Therefore, the outcome of the construct domain consolidation of measures of competence was the identification of 12 domains. There were significant differences in the predominant domains assessed in IM versus RM studies, with less focus on professionalism and knowledge-based domains in the latter.

Instrument Properties

Instrument properties were evaluated for their attention to the reporting of measurement properties such as reliability, validity, and interpretability. Despite the preponderance of identified measures, there is a dearth of evidence for their reliability and validity; 53.5% of the identified measures provided reliability and validity data (37/73 IM and 45/80 RM), whereas 21.5% of studies described measures without reference to either their reliability or validity (16/73 IM and 17/80 RM).

Measurement content validity was predominantly based upon citation in several of the measures[e.g.M:46,47,48,50,52,64,65,114] rather than validation against an agreed standard measure. With regard to internal consistency, a Cronbach's alpha (α) value was reported for 39 reviewed measures. Cross-cultural validity was addressed in 29.4% of measures. Predictive validity was addressed by a small number of studies to measure the predictive value of pre-medical school

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assessment on later performance, and the predictive value of cognitive assessment in predicting subsequent impaired competence in aging doctors[P:81,M:123]. Computerised, written, and oral measures were typically single-domain measures of knowledge but reported higher levels of validity.

Details on measurement interpretability were found for 119 reviewed measures. As many established measurement methods were institution-specific there was substantial variability in testing procedures, content domains and subsequent outcomes. However, practice and test-retest effects in longitudinal studies received minimal consideration[P:42, 81]. Testing measures of competence in real practice using double-blind observations are evidently unfeasible for logistical and ethical reasons, however workplace-based 360-degree feedback from a doctor or trainees colleagues provides an alternative whereby varied sources of in-practice observations with real patients and there is evidence for the validity of this method[M:33,43].

Measurement Assessment Context

Increasing effort has been dedicated to enhancing the realism of technical skills training by introducing SP's and role-plays[P:51]; 45.4% of the reported measures (53/73 IM and 30/80 RM) were designed to be assessed in a simulated context, and just 20.9% with real patients (only 18 of which were IM), 33.7% of measures used neither. The majority assessed competence in either a workplace-based assessment (n = 19), or (more commonly) a simulated assessment (n = 67). However, the acceptability and feasibility of real patient measures rather than simulated context assessment encounters has significant cost, time and ethical barriers. However, there has been increased emphasis in more recent studies on assessments which attempt to mimic real world clinical situations.

Positional Orientations

Studies assessed competence measurement from two positional orientations; assessor and assessee. Measurement of assessee competence was categorised by four populations; medical school applicants (n = 3), medical students (n = 60), junior doctors (n = 19) and fully trained (senior) doctors (n = 20). The assessment of medical school applicants for core

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competencies received minimal attention (3 studies) in the selected literature; one research study[P:9] utilized a video-based situational judgement test (SJT)[M:16] assessing social competencies, the second[P:58] addressed an assessment framework measure[M:91], and the third[P:60] employed a coping responses inventory[M:94] in addition to a state entrance examination[M:95]. Medical students were the target population in the majority of studies, as a competency-based curriculum is becoming a dominant organizing framework for medical education and assessment[P:47], and its emphasis was reflected in its research dominance in the literature. Only three studies examined multiple assessee populations; one[P:60] studied medical school applicants, medical students, junior doctors, and fully qualified doctors using psychometric burnout[M:94] and knowledge test[M:95] measures, and found that the assessment of coping competence at medical school enrolment was a predictor of longitudinal professional competence and career development. The second[P:72] (medical students, junior doctors, and fully qualified doctors) used a group evaluation[M:113] to explore the contextual-environmental factors that affect competence assessment, and the third study[P:92] found that vignettes[M:135], SP's[M:136], and chart abstraction[M:137] measures were a valid and comprehensive method to measure competence in junior doctors and fully qualified doctors.

Assessor perspectives are important in the interpretation of competence outcomes. Assessor-performed analogue rating scales were used in 47 measures (31 IM and 16 RM). Evaluative standpoints of assessors varied between the selected studies and were described as follows: Fifty measures (54.8% of IM, 12.5% of RM) employed supervisory occupational appraisal of a junior or student. Six measures (4.1% of IM, 3.8% of RM) used peer assessment. Twenty-two measures (19.2% of IM, 10% of RM) utilised patients or SP's who can provide a key role in evaluating competence as judged by their direct experience of the doctor or trainee. Four measures (2.7% of IM, 2.5% of RM) used disciplinary/regulatory assessment procedures designed to assess a comprehensive range of domains. Finally, 37 measures (8.2% of IM, 38.8% of RM) involved self-assessment of competence. Therefore, a diverse range of assessor standpoints influence competence assessment and providing feedback is arguably[P:23] a complex affective process and the assessment is determined by the assessor's cognition, beliefs, and emotions. Inter-rater reliability is pertinent to the interpretability of selected measures in this review as the majority of multi-domain IM evaluations of clinical performance, such as OSCE's, and the mini-CEX rely extensively on rater-assessment. Inter-rater reliability was reported for a small number of studies[e.g.P:51,67,75,77,89], however many were monocentric studies.

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Therefore, many assessments of competence are contingent upon assessor subjectivity. Whilst some studies demonstrated attempts to assess inter-rater reliability, this is an important issue that requires further attention.

DISCUSSION

This systematic review sought to identify and evaluate measures of competence in doctors and medical students. 105 articles were identified that included a broad range of 153 competence assessment measures. The principal findings were as follows: Very few studies conceptualised competence through definition or operationalising the construct. Therefore, although competence is considered paramount for safe practice, no consensus in the selected literature exists regarding its conceptualisation, nor is there measurement standardisation. Twelve domains used to conceptualise and measure competence were identified. The deconstruction of each domain and its relevance yielded strong operational purpose in 11 of the domains but identified inconsistencies and procedural subjectivities regarding Professionalism (Domain 5). Assessment of professionalism in the studies lacked reliable measurement evaluation tools and was open to implicit judgment biases; this review therefore questions whether further objective assessment of professionalism may be identified and validated. In contrast, judgement and bias (Domain 3) included a range of specific measures applied to assess competence with diversity and judgement orientations; the findings that patient characteristics were commonly associated with treatment outcomes requires continuous examination from a training and assessment perspective to ensure non-discriminatory ethical treatment of all patients[10]. However, it is important to acknowledge that the competence domain categories are not necessarily indicative or proportionally reflective of all globally accepted assessment tools. There are likely varying conceptualisations of competence in different cultures, and in different clinical, social and economic contexts, therefore further research on geographic and cultural construct analysis is advocated to explore potential differences in worldwide assessments.

Interestingly, the distribution of competence domains differed between IM and RM; the RM reflecting more explicit and focused consideration of the contribution of behaviouristic

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(skills and performance), systemic pressures, psychophysiological (cognitive, emotional, health), and vantage-point (beliefs, perceptions and attitudes) factors as conceptually relevant to competence. A further follow-up exploration to examine which of the RM are being implemented in systemic operational practice would be worthwhile. It is noteworthy that findings related to the validation of the measures were often limited or entirely absent, therefore further validation of existing measures is recommended. There is a broadly held assumption that assessment formats with singular competence domains are limited[11] whereas utilising comprehensive multidimensional assessment measures will minimise deficiencies[12]. Furthermore, the majority (n = 76) of selected studies were single-centre assessments. To address this, we encourage multicentric testing of competence assessment to offer greater generalisability of the data. Correspondingly, the sharing of assessment measures between institutions may encourage greater homogeneity in practices. There has been increasing interest recently in programmatic longitudinal assessment of both students and doctors using multiple assessment techniques, in which individual assessment elements are part of a process of gathering information contributing to the overall assessment rather than being used on their own to make decisions. In this model these assessments contribute to the learning process, breaking down the traditional dichotomy between formative and summative assessment [13].

A wide range of global competence measures were reported in the selected studies. These included written assessments, OSCEs, anatomic simulations, and workplace-based assessments. The combination of a written test with a clinical skill OSCE test has been demonstrated to increase the predictive validity of either alone. In research studies assessing competence there is considerable interest in simulation-based assessments and some have focussed on cognitive assessment which may be of particular importance with many doctors working beyond retirement.

Assessor bias is a worthwhile area of further research due to concerning discrepancies in assessment accuracy that have been attributed to unreliable self-assessment,[14] inconsistent applications of organizational standards resulting in a lack of reliability, and unclear rating scale criteria. Assessments involving direct observations inherently include automatic judgement processes, and don't necessarily provide objective assessment,[15]. However, it has been argued,[16] that standardisation may be a constraint on authentic assessment interpretations and delineations. Approaches using 360-degree global rating evaluations by multidisciplinary multilevel colleagues who interact with a doctor or trainee routinely may

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improve objectivity through a comprehensive range of perspectives compared to single-assessor perspectives,[17]. This review highlighted the inconsistent judgement acuity of assessors as paramount to competence assessment[e.g.P:1,5,7,17,18,20,28,36,47,52,69,101]. When evaluating instrument properties, the consideration of feasibility is paramount, as all measures require resources for systemic implementation. Furthermore, the number of individuals being assessed, the time constraints, financial resources, the interpretability, the validity and reliability of measures are key considerations. When reviewing these points, many experimental measures are unfeasible as routine in-practice measures due to resource availability and/or ethical and time constraints.

There was unanimity in the literature that competence assessment throughout a doctor's career is necessary to protect patients and the profession,[1]. Therefore, continued exploration of evidence-based measures to assess competence throughout doctors' training and career is advocated. This review highlights the absence of consensus on the domains relevant to competence, and a lack of consistency in existing clinical measurement standards. In conclusion, the findings of this study strongly support the use of multidomain evidence-based measurement-selection, and attending to developing measures that replicate the unique systemic context through the evaluation of all domains relevant to competence is recommended as a continued focus of research in this area.

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Contributors All authors developed the protocol for this systematic review. KH conducted the screening of studies, data extraction and critical appraisal. DL and AL reviewed each stage of study selection and calibrated findings. All authors assisted in the interpretation and write up of results. All authors approved the final manuscript prior to submission.

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Ethics Approval Institutional review board clearance is not required for this systematic review as all data are publicly available

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FIGURE LEGENDS

Figure 1 reports the flow diagram of the review search strategy and study selection based on PRISMA (2009) guidelines

Figure 2 presents the percentage distribution of all articles' geographic origin

Figure 3 reports the distribution of domains of competence assessed in selected studies reporting in-practice assessments of competence, and those from research studies

Figure 4 presents donut charts representing the percentage of the procedures that were used in the measures within each of the twelve domains

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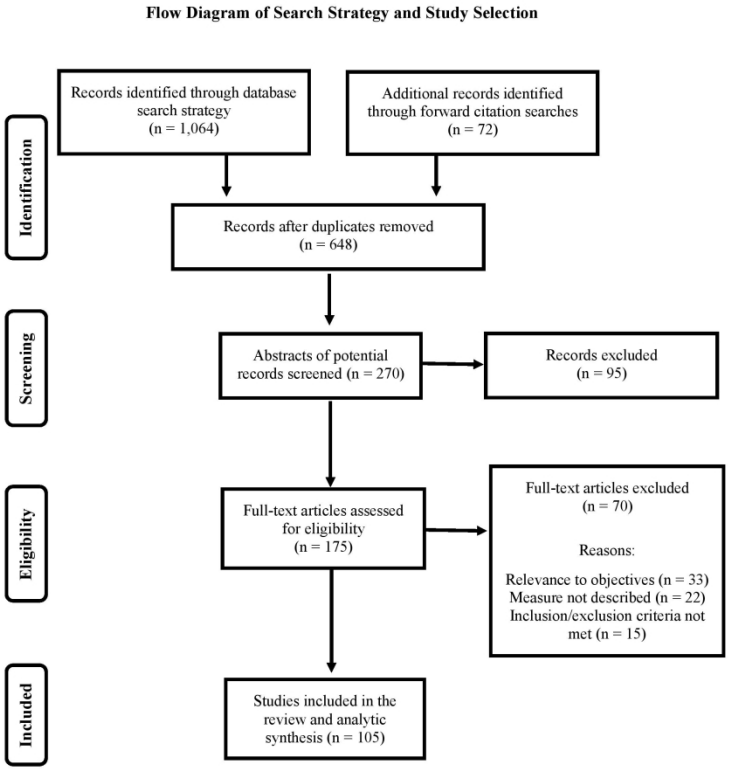


Figure 1 Flow Diagram of Search Strategy and Study Selection Based on PRISMA (2009) Guidelines

Figure 1 reports the flow diagram of the review search strategy and study selection based on PRISMA (2009) guidelines

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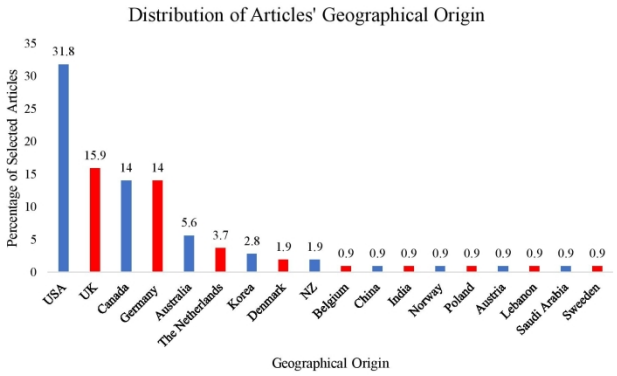


Figure 2 The Percentage Distribution of Selected Articles' Geographical Origin

Figure 2 presents the percentage distribution of all articles' geographic origin.

209x297mm (300 x 300 DPI)

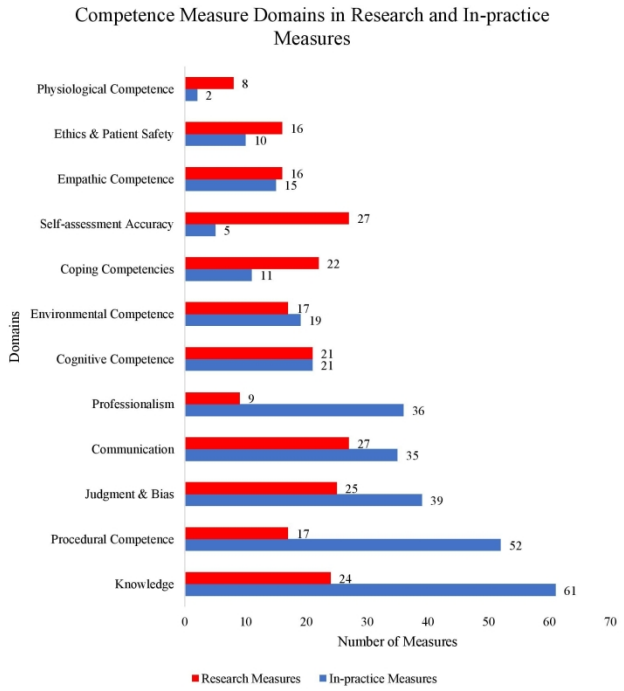


Figure 3 Competence Domains in Research and In-practice Measures

Figure 3 reports the distribution of domains of competence assessed in selected studies reporting in-practice assessments of competence, and those from research studies

209x297mm (300 x 300 DPI)



Figure 4 Donut charts representing the distribution of procedural variation within each competence domain

Figure 4 presents donut charts representing the percentage of the procedures that were used in the measures within each of the twelve domains

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MEASURING COMPETENCE IN DOCTORS

Supplemental Material File 1

Study Search and Selection Results

Electronic Search Engine		Total Identified Records	Number Following Deduplication	Relevant Titles Selected for Abstract Screening	Full Text Articles Assessed for Eligibility	Studies Included in Analytic Synthesis
PsychSource	English Language Search	221	221	31	17	7
	American-English Search	224	42	12	9	6
Medline/ PubMed	English Language Search	132	65	35	19	10
	American-English Search	179	57	39	14	9
Cochrane Library – Central	English Language Search	53	41	20	19	13
	American-English Search	117	56	17	18	11
Web of Science	English Language Search	58	52	19	14	10
	American-English Search	80	42	25	15	11
Specific Paper Retrieval at Journal Site – Forward Citation Search		72	72	72	51	28
Total		1, 136	648	270	175	105

Note. Search results are reported in the sequence they were undertaken

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Supplemental Material File 2

Assigned Reference Identifier Codes and Full References

Identifier	Reference	Identifier	Reference
P1	Boursicot (2006)	P46	ALQahtani *et al.*, (2018)
P2	Hojat *et al.*, (2002)	P47	Yeates *et al.*, (2012)
P3	Ogle *et al.*, (2013)	P48	Frey *et al.*, (2002)
P4	Maudsley *et al.*, (2007)	P49	Huizinga *et al.*, (2018)
P5	Wilkinson & Frampton (2003)	P50	Aper *et al.*, (2012)
P6	Williams & Klamen (2015)	P51	Nikendei *et al.*, (2007)
P7	Morgan & Cleave-Hogg (2002)	P52	Hall *et al.*, (2015)
P8	Burgess *et al.*, (2011)	P53	Truncali *et al.*, (2011)
P9	Frölich *et al.*, (2017)	P54	Chumley *et al.*, (2006)
P10	Dahlin *et al.*, (2010)	P55	Ludwig *et al.*, (2018)
P11	Kell *et al.*, (2014)	P56	Windish *et al.*, (2005)
P12	Wilkinson & Frampton (2004)	P57	Bugaj *et al.*, (2018)
P13	Oh *et al.*, (2016)	P58	Schreurs *et al.*, (2018)
P14	Homberg *et al.*, (2019)	P59	Gude *et al.*, (2017)
P15	Monteiro *et al.*, (2019)	P60	Tartas *et al.*, (2016)
P16	Radabaugh *et al.*, (2019)	P61	Han & Chung (2016)
P17	Harendza *et al.*, (2019)	P62	Devine *et al.*, (2015)
P18	Madrazo *et al.*, (2018)	P63	Roland *et al.*, (2015)
P19	Woods *et al.*, (2018)	P64	Carr *et al.*, (2014)
P20	Prediger *et al.*, (2019)	P65	Mehdizadeh *et al.*, (2014)
P21	Gerard *et al.*, (2018)	P66	Ammentorp *et al.*, (2013)
P22	Renaud & Côté (2017)	P67	Terregino *et al.*, (2019)
P23	Sehlbach (2020)	P68	Zelin *et al.*, (2018)
P24	Olsen *et al.*, (2019)	P69	Pool *et al.*, (2018)
P25	Burnett *et al.*, (2019)	P70	Patterson *et al.*, (2017)
P26	Petrosoniak *et al.*, (2019)	P71	Zabar *et al.*, (2016)
P27	Siau *et al.*, (2019)	P72	Stroud *et al.* (2015)
P28	Kain *et al.*, (2019)	P73	Park *et al.*, (2015)
P29	Prediger *et al.*, (2020)	P74	Dong *et al.*, (2015)
P30	Giesler *et al.*, (2011)	P75	Ginsburg *et al.*, (2015)
P31	Bodamer *et al.*, (2015)	P76	Ferenchick & Solomon (2013)
P32	Beckman *et al.*, (2012)	P77	Thomas *et al.*, (2011)
P33	Hautz *et al.*, (2017)	P78	Ginsburg *et al.*, (2010)
P34	Krautter *et al.*, (2011)	P79	Jani *et al.*, (2017)
P35	Liddell *et al.*, (2002)	P80	Spittal *et al.*, (2015)
P36	Yazbeck-Karam *et al.*, (2014)	P81	Turnbull *et al.*, (2006)
P37	Engerer *et al.*, (2016)	P82	Caulford *et al.*, (1994)
P38	Posner *et al.*, (2013)	P83	Grace *et al.*, (2014)
P39	Mc Evoy *et al.*, (2016)	P84	Lehmann *et al.*, (1990)
P40	Snow *et al.*, (2016)	P85	Meagher *et al.*, (2009)
P41	Hoefer *et al.*, (2017)	P86	Mehdizadeh *et al.*, (2014)
P42	Shapiro *et al.*, (2009)	P87	Unwin *et al.*, (2019)
P43	Krautter *et al.*, (2014)	P88	Devitt *et al.*, (1998)
P44	Stroben *et al.*, (2016)	P89	Joshi *et al.*, (2004)
P45	Christiansen *et al.*, (2014)	P90	Smith *et al.*, (2019)

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Identifier	Reference	Identifier	Reference
P91	Santen *et al.*, (2013)	P99	Green *et al.*, (2017)
P92	Peabody *et al.*, (2000)	P100	Mitchell *et al.*, (2011)
P93	Humphrey *et al.*, (2011)	P101	Tham *et al.*, (2017)
P94	Gollehon *et al.*, (2017)	P102	Shulruf *et al.*, (2019)
P95	Austin *et al.*, (2007)	P103	Shi *et al.*, (2018)
P96	Kirby *et al.*, (2011)	P104	Trockel *et al.*, (2018)
P97	Wang *et al.*, (2018)	P105	West *et al.*, (2006)
P98	Sevlever & Rice (2010)		

Selected Paper Review References

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MEASURING COMPETENCE IN DOCTORS

Supplemental Material File 3

Comprehensive List of All Identified Measures

Reference Identifier	IM/ RM	Outcome Measure	Measure Identifier
P1	IM (A)	The Angoff Method (OSCE rating scale)	M1
P2	IM	US Medical Licensing Examinations (USMLE)	M2
P2	IM	Medical College Admission Test (MCAT)	M3
P2	RM	Jefferson Scale of Physician Empathy	M4
P3	RM	Jefferson Scale of Physician Empathy – Student Version (JSPF-S)	M5
P3	IM	Objective structured clinical examination (OSCE)	M6
P3	IM	Rating Scales for the Assessment of Empathic Communication in Medical Interviews (REM)	M7
P4	RM	Questionnaire survey about notions of a 'good doctor' 6 brief questionnaire surveys	M8
P5	IM (A)	Performance Rating Instrument - A Trainee Intern Rating Form	M9
P6	RM	Twenty Questions (TQ) game performance	M10
P6	IM	Medical College Admission Test (MCAT) Biological Science Test Score	M11
P7	RM	25-point prior experience and confidence questionnaire	M12
P7	IM	A pre-determined critical event simulation	M13
P8	RM	Vignettes on chronic pain (assessed by treatment outcome)	M14
P8	IM	The short Medical Authoritarianism Scale (MA-S)	M15
P9	RM	A video-based situational judgment test (SJT) assessing social competencies	M16
P10	RM	Performance-based self-esteem (PBSE)	M17
P10	RM	Oldenburg Burnout Inventory (OLBI)	M18
P11	IM	Standardised Patient Examination (SPE)	M19
P11	IM	Opinions about Physicians Interactions with Patients (OPIP)	M20
P11	IM	Medical College Admission Test (MCAT) GPA	M21
P12	IM	OSCE combined with a written examination	M22
P13	RM	Anderson's Epistemological Beliefs Questionnaire (EBQ)	M23
P13	IM	SP: Korean Clinical Performance Examination (CPX)	M24
P14	IM (A)	The CanMEDS Physician Competency Framework	M25
P15	IM	Written Vignettes	M26
P15	RM	Time-pressured Vignette Assessment	M27
P16	IM	US Medical Licensing Examinations (USMLE)	M28
P17	RM	Group Assessment Performance (GAP) Test	M29
P18	IM	OSCE self-assessment measure on a 6-point Global Rating Scale (GRS)	M30

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P19	RM	Workload InsightTM Application (Chronometry measure of task)	M31
P19	IM	The Secondary Task Method – Venipuncture (Simulated task performance rated using a 16-point checklist)	M32
P20	RM (A)	A 360-degree Assessment of Competencies	M33
P21	RM	PediatricSim Electronic Game	M34
P22	RM	The 5-item Doctor-Patient Communication Competency Instrument for Medical Students (SP)	M35
P23	RM	Patient Evaluation of Physician Competence	M36
P24	RM	Diagnostic Competency Framework	M37
P25	IM	Simulated Patient (SP) with simulated crisis parameters of mortality and morbidity	M38
P26	IM	7-point Global Rating Scale (GRS)	M39
P26	RM	Chronometry	M40
P27	IM	Direct Observation of Procedural Skills (DOPS)	M41
P28	RM (A)	Qualitative Experiential Knowledge	M42
P29	RM	360-degree Assessment	M43
P30	RM	The Freiberg Questionnaire to Assess Competencies in Medicine	M44
P31	RM	Internal Medicine Standardized Simulation Based Examination (SSBE)	M45
P32	RM	The Interpersonal Reactivity Index (IRI)	M46
P32	RM	The Maslach Burnout Inventory	M47
P32	RM	Medical Outcomes Study Short Form (SF-8)	M48
P32	IM	The Mini-Clinical Evaluation Exercise (mini-CEX)	M49
P32	IM	In-training Examinations (ITE)	M50
P33	IM	Global Rating Scale (GRS)	M51
P33	RM	The Test of Self-Conscious Affect, Version 3 (TOSCA-3)	M52
P34	RM	Chronometry	M53
P34	RM	Objective Performance Ratings by Independent Assessors of Videographed Procedure	M54
P35	IM	Procedural Assessment of Clinical Skill (Injection & Suturing)	M55
P35	RM	Procedural Skills Confidence Survey	M56
P36	IM	OSCE (SP)	M57
P37	IM	Student Self-Ratings of Communication Competence (VAS)	M58
P38	IM	SP-PTT (part-task trainer) hybrid: OSCE	M59
P38	IM	PTT (part-task trainer): OSCE	M60
P39	RM	Written reflective assignment	M61
P39	RM	Rating Scale - self assessment	M62
P40	IM	OSCE	M63
P40	IM	Doctors' Interpersonal Skills Questionnaire (DISQ)	M64
P41	IM	OSCE Station (videotaped)	M65
P42	RM	Self-Assessment of Interpersonal Competence Questionnaire (SAICQ)	M66
P42	RM	The Staff-Patient Interaction Scale (SPIR)	M67
P42	RM	Interpersonal Skills Rating Scale (ISRS) rated by SP's	M68

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P43	RM	Self efficacy rating (Likert scale)	M69
P43	IM	SP clinical performance assessment by assessor direct observation (DO)	M70
P44	RM	Self-reported preparedness (self efficacy) Likert questionnaire	M71
P44	IM	SP observation using the Calgary-Cambridge Observation Guide (CCOG)	M72
P45	IM	Radiological (X-ray) diagnostic accuracy	M73
P45	RM	Diagnostic Confidence Rating (Likert scale)	M74
P46	RM	Diagnosis using written clinical vignettes on Qualtrics	M75
P46	RM	Chronometry	M76
P47	IM (A)	mini-CEX	M77
P48	RM	Computer-analyzed Electroencephalography	M78
P48	RM	Pauli Test	M79
P49	RM	The Mini-NeuroCart	M80
P50	RM	Self efficacy scale	M81
P50	RM	Responses to a standardised video case	M82
P51	IM	OSCE test - Global rating scale of communication skills - binary checklist	M83
P52	RM	Analogue Patient Satisfaction Questionnaire	M84
P53	IM	OSCE SP checklist	M85
P54	IM	SP case	M86
P55	IM	Formative exit exam (medical school module) on clinical reasoning and diagnostic accuracy	M87
P56	IM	SP's who rated students on a 5-point scale for communication skills	M88
P56	IM	Diagnostic Thinking Inventory	M89
P57	IM	Procedural Protocol Instrument (IPPI) Global rating scale - blind video assessed	M90
P58	RM	CanMEDS Assessment Framework Measure	M91
P59	RM	The Oslo Inventory of Self-Reported Communication Skills (OSISCS)	M92
P59	IM	SP - The Arizona Communication Interview Rating Scale (ACIR)	M93
P60	RM	Coping Responses Inventory (Moos)	M94
P60	IM	State Examination for Medical Doctors	M95
P60	RM	Maslach Burnout Inventory (MBI)	M96
P61	IM	CPX - Clinical Performance Examination (SP)	M97
P61	IM	OSCE	M98
P62	IM	MRCP (UK)	M99
P62	IM	MRCGP AKT (Applied Knowledge Test)	M100
P63	RM	Self-evaluation meta-planning focus group	M101
P64	IM	The Junior Doctor Assessment Tool (JDAT)	M102
P65	IM	GMC test of Competence (ToC)	M103
P65	IM	OSCE	M104
P65	RM	Self-Assessment Questionnaire - ranking in comparison to other doctors	M105
P66	IM	OSCE - using the Calgary-Cambridge Observation Guide Checklist (SP and observer)	M106

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P66	RM	Self-assessment - Likert scale	M107
P67	RM	Empathy and Clarity Rating Scale (ECRS) in OSCE evaluation	M108
P68	RM	Comfort and competence of working with sexual and gender minorities survey	M109
P69	IM (A)	Portfolio-based assessment	M110
P70	RM	Wechsler Adult Intelligence Scale (WAIS-IV)	M111
P71	RM	OSCE - Standardized nurse (SN) rating on professional collaboration	M112
P72	RM	Focus groups	M113
P73	IM	OSCE - 4 stations (SP)	M114
P73	RM	5-minute examination - Diagnostic Accuracy	M115
P74	RM	The Essential Elements of Communication (EEC) - rating scale on an OSCE (SP)	M116
P75	RM	Socio-cultural domains of patient safety - OSCE (SP)	M117
P76	RM	Electronic 'Just in Time Medicine' (JIT) clinical evaluation exercize (CEX)	M118
P77	IM	Group Assessment using faculty opinion survey	M119
P78	IM (A)	Assessor Rating Scale	M120
P79	IM	360-degree feedback	M121
P80	RM	PRONE score	M122
P81	RM	PREP assessment (Physician Review Program) Test Battery	M123
P82	RM	Chart stimulated recall (CSR)	M124
P82	IM	SP's who rated students on a 5-point scale for communication skills	M125
P83	IM	Center for Personalized Education for Physicians (CPEP) multivariate competence assessment	M126
P84	RM	Patient-physician Interaction Quality Checklist	M127
P85	RM	Team Objective Structured Bedside Assessment (TOSBA)	M128
P86	IM	GMC Test of Competence (ToC) - including a written knowledge test and an OSCE	M129
P87	IM	GMC Test of Competence (ToC) - including a written knowledge test and an OSCE	M130
P88	RM	Knowledge-based questionnaire	M131
P89	RM	360-degree Assessment	M132
P90	IM	OSCE (SP)	M133
P91	RM	Emergency Medicine Milestones Survey	M134
P92	IM	Vignettes	M135
P92	IM	SP's	M136
P92	IM	Chart abstraction	M137
P93	IM	Referral for fitness to practice investigation	M138
P94	IM	OSCE	M139
P95	RM	The Jefferson Scale of Physician Empathy (JSE)	M140
P96	RM	The Scale of Attitudes Toward Disabled Persons (SADP)	M141
P97	RM	The Jefferson Scale of Physician Empathy (JSE)	M142
P98	RM	The Almost Perfect Scale-Revised (APS-R)	M143
P98	IM	Grade Point Average (GPA)	M144

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P99	RM	The cross-cultural care survey (CCCS)	M145
P100	IM	Workplace Based Assessments (WPBA) including: mini-clinical evaluation exercize (mini-CEX), case-based discussion (CbD), direct observation of procedural skills (DOPS), and mini-peer assessment tool (mini-PAT)	M146
P101	IM (A)	Workplace Based Assessments (WBA's)	M147
P102	IM	OSCE	M148
P103	RM	Support vector machine combined with an ordinal partitioning model (SVMOP)	M149
P104	RM	Professional Fulfilment Index (PFI)	M150
P105	RM	Epworth Sleepiness Scale	M151
P105	RM	QOL Burnout and depression	M152
P105	RM	Self-reported medical errors	M153

Notes.

EM: Existing Measures, RM: Research Methods, (A): Assessor assessment.

Papers in the manuscript are referenced in the format $^{P1,2,3,4,5\ etc.}$ individual measures are referred to as ^{M1,2,3,4,5} etc.

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MEASURING COMPETENCE IN DOCTORS

Supplemental Material File 4

List of Measures in Each Domain

Domain 1				Domain 2				Domain 3			
M1	M33	M72	M117	M2	M41	M88	M129	M2	M51	M100	M147
M2	M34	M73	M118	M6	M43	M90	M130	M4	M52	M102	
M6	M36	M75	M119	M8	M45	M93	M134	M5	M57	M103	
M8	M37	M82	M124	M9	M49	M95	M136	M6	M59	M104	
M9	M38	M83	M125	M13	M50	M97	M139	M8	M60	M106	
M10	M39	M85	M126	M14	M51	M98	M148	M9	M63	M109	
M11	M41	M86	M128	M19	M54	M99		M10	M65	M111	
M12	M43	M87	M129	M20	M55	M100		M13	M70	M114	
M13	M44	M88	M130	M22	M56	M102		M14	M72	M115	
M14	M45	M89	M131	M24	M57	M103		M15	M73	M118	
M17	M49	M90	M133	M25	M59	M104		M19	M74	M119	
M19	M50	M95	M134	M28	M60	M106		M28	M75	M126	
M21	M51	M97	M135	M29	M63	M114		M29	M77	M129	
M22	M54	M98	M136	M30	M65	M115		M31	M84	M130	
M24	M55	M99	M137	M32	M70	M117		M33	M87	M135	
M25	M57	M100	M139	M33	M72	M119		M34	M88	M136	
M26	M59	M102	M144	M34	M73	M122		M36	M90	M139	
M27	M60	M106	M146	M36	M77	M124		M37	M91	M141	
M28	M63	M109	M147	M37	M83	M125		M38	M97	M143	
M30	M65	M114	M148	M38	M85	M126		M43	M98	M145	
M32	M70	M115		M39	M86	M128		M45	M99	M146	

Domain 4			Domain 5			Domain 6		Domain 7		Domain 8	
M8	M67	M112	M1	M98	M139	M3	M75	M8	M75	M18	M94
M9	M68	M116	M22	M99	M148	M8	M79	M13	M76	M27	M96
M16	M83	M117	M33	M100	M150	M9	M80	M18	M90	M29	M103
M20	M84	M118	M36	M102		M10	M82	M19	M91	M31	M104
M25	M85	M119	M38	M103		M11	M89	M25	M103	M32	M113
M29	M86	M121	M41	M104		M21	M91	M27	M104	M33	M129
M33	M88	M125	M43	M106		M22	M99	M29	M113	M38	M130
M35	M92	M126	M44	M114		M23	M100	M31	M117	M43	M134
M36	M93	M127	M51	M115		M24	M103	M32	M119	M44	M139
M37	M95	M128	M52	M117		M26	M104	M33	M126	M47	M143
M38	M97	M129	M59	M119		M27	M111	M37	M129	M48	M150
M41	M98	M130	M60	M121		M29	M117	M38	M130	M51	M152
M43	M99	M132	M63	M125		M31	M118	M40	M133	M52	
M44	M100	M133	M65	M126		M32	M123	M43	M136	M56	
M46	M102	M136	M70	M128		M33	M128	M47	M139	M59	
M52	M103	M139	M72	M129		M34	M129	M51		M60	
M58	M104	M146	M83	M130		M37	M130	M52		M71	
M59	M106	M147	M85	M133		M38	M139	M53		M75	
M60	M108	M148	M90	M136		M41	M144	M59		M80	
M64	M109	M149	M95	M137		M43	M146	M60		M90	
M66	M111		M97	M138		M44	M147	M72		M91	

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Domain 9		Domain 10		Domain 11		Domain 12
M8	M96	M4	M121	M1	M138	M8
M12	M101	M5	M126	M4	M146	M18
M17	M105	M7	M129	M5	M147	M31
M18	M107	M8	M130	M8	M150	M48
M23	M113	M9	M132	M9	M153	M78
M25	M124	M14	M136	M14		M80
M30	M126	M15	M140	M15		M129
M33	M129	M36	M142	M23		M130
M43	M130	M43	M146	M36		M150
M52	M134	M46	M147	M43		M151
M56	M143	M58		M52		
M61		M59		M61		
M62		M60		M62		
M66		M61		M91		
M69		M62		M117		
M71		M64		M121		
M74		M84		M122		
M80		M91		M123		
M81		M106		M126		
M92		M108		M129		
M94		M109		M130		

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6-7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7-8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8-9

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Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8-9
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-15
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10-19
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	20-22
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	21
FUNDING			

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Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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The variable trajectory: A systematic review, analytic synthesis and construct domain consolidation of international measures of competence in doctors and medical students.

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THE VARIABLE TRAJECTORY: A SYSTEMATIC REVIEW, ANALYTIC SYNTHESIS AND CONSTRUCT DOMAIN CONSOLIDATION OF INTERNATIONAL MEASURES OF COMPETENCE IN DOCTORS AND MEDICAL STUDENTS

Running title: Measuring competence in doctors

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The variable trajectory:
A systematic review, analytic synthesis and construct domain consolidation of international measures of competence in doctors and medical students

ABSTRACT

Background

Competence is assessed throughout a doctor's career. Failure to identify and manage impaired competence can have critical consequences. Consistent conceptualisation and accurate measurement of this construct is imperative. Therefore, the objective of this review was to identify and evaluate measures used to assess competence in doctors and medical students.

Methods

A systematic search of the published literature was undertaken between December 2019 and February 2020 for articles reporting on the measurement of competence in doctors and/or medical students. Searches were conducted in the PsychSOURCE, US National Library of Medicine National Institutes of Health, MEDLINE (PubMed), The Cochrane Central Register of Controlled Trials (CENTRAL), and Web of Science electronic databases. Citation screening and forward citation tracking of included studies was carried out to identify any further relevant papers for inclusion. One thousand one-hundred and thirty-six potentially relevant articles were screened. An analytic synthesis approach was implemented to the identification, organization and interpretation of homogenous study and measure characteristics.

Results

Twelve competence domains were identified from the 153 identified measures. Knowledge and Procedural competence domains were the dominant focus of publications reporting current medical practice, but less so in research-based studies which more frequently assessed interpersonal, psychological, physiological and ethical competencies. In the 105 included articles,

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the reporting of measurement instrument quality was varied, with comprehensive reporting only present in 53.6% of measures; validation for some of the measures was particularly limited.

Discussion

Whilst this review included a considerable number of publications reporting the measurement of competence in doctors and medical students, the heterogeneity of the measures and variation of findings limit the ability to evaluate their validity and generalisability. However, this review presents a resource for researchers and medical educators which may inform operational practice and future research.

Keywords: Competence · Doctor · Medical student · Physician · Performance

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Strengths and limitations of this review

- To our knowledge, this is the first contemporary systematic review to consolidate globally recognised domains relevant to competence assessment of doctors and medical students in both research and in-practice measures.

- This PROSPERO registered review employed an extensive and rigorous search strategy of four databases and forward citation searches using defined selection and abstraction criteria calibrated by three researchers, undertaken in alignment with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement, and guided by the synthesis without meta-analysis (SWiM) reporting guidelines.

- The review identified a broad range of measures used to assess competence, and identifies the homogeneity and heterogeneities using a textual and conceptual analytic synthesis approach.

- A limitation of this review is that English language restrictions were placed on the searches, and although research studies were derived from 18 countries, the inclusion of further languages may yield unidentified studies, and subsequent differences in worldwide assessments of competence.

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INTRODUCTION

Competence is assessed throughout doctors' professional lifespan, from selection for entry to medical school to medical student to junior doctor, to senior doctor until retirement. The timely identification, support and remediation of doctors who do not demonstrate sufficient competence to undertake their duties safely is imperative. However, the proportion of doctors who are incompetent, or dyscompetent, and failing to uphold adequate practice standards is unclear[1]. Robust assessment of competence necessitates accurate and standardised measures that comprehensively examine all domains relevant to the core competencies required to undertake a job effectively. It has been argued that competencies are learned, complex context-specific dispositions, but the complexity and diversity inherent in doctors' responsibilities make measurement of competence a complex task[2].

Despite numerous definitions,[3] an agreed explicit, amalgamated, and quantifiable definition of the construct of competence, and in many cases a consistent and transparent chain of inferences to validate the observed assessment and measurement of competence are lacking[4]. It has been argued,[5] that it is challenging to identify reliable and valid measures of competence as many of the skills are considered difficult to quantify. Contemporary competence-based medical education is predominantly driven by patient safety, efficiency and clinical outcomes[6].

In order to explore these challenges, and provide an overview of the diverse facets of competence, and their dominance within current assessment measures; the objective of this systematic review of peer-reviewed scientific literature was to identify and evaluate specific measures of competence relevant to doctors and medical students. As the accurate measurement of competence is paramount to patient safety, a systematic review of the measures employed and domains of competence assessed in medical student selection, medical students and doctors was considered important.

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METHOD

The review protocol was registered with PROSPERO, adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement,[7] and guided by the 'synthesis without meta-analysis (SWiM) reporting guidelines'[8].

Literature Search

Data Sources

A systematic search of the literature was conducted in the PsychSOURCE, US National Library of Medicine National Institutes of Health, MEDLINE (PubMed), The Cochrane Central Register of Controlled Trials (CENTRAL), and Web of Science electronic journal databases. Citation screening and forward citation tracking of included studies was carried out to identify any further relevant papers; these were retrieved from the journal site or academic library databases.

Search Strategy

A pilot search was undertaken to determine relevant electronic search engines, refine the search strategy, define search keywords, identify relevant MeSH terms, determine search restrictions, and select Boolean operators. English and American-English language terms were used. Search terms employed the keywords; 'doctor', 'competence', and 'performance', incorporating the Boolean operator ('and'), in addition to MeSH terms (physicians, medical students). Restrictions on the databases included human studies, published in English language, and full text availability. No date restrictions were applied. Searches were undertaken between December 2019 - February 2020. Further searches in response to the literature were performed through reference lists of reviewed articles in forward citation searches. Results were downloaded to reference management software and de-duplicated.

Study Identification and Selection

Inclusion/Exclusion Criteria

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Studies were included if they described the measurement of competence in medical students and doctors, with full-text availability in peer-reviewed journals, published in English language. Commentaries, editorials, letters, book chapters, conference papers, books, doctoral theses, dissertations, and articles which did not adequately report competence assessment measures were excluded.

Calibration of Findings

One reviewer undertook title and abstract screening, and selection methods were supervised by two reviewers who verified the extraction of data, checked consistency and methodological clarity. Study selection disagreements and subsequent queries about inclusion following full-text review were resolved by discussion between researchers to reach consensus.

Study Selection

From searches, 1,136 papers were identified; after duplicates were removed 648 original articles remained. Following assessment for relevance to the objectives and selection criteria 105 articles were selected for systematic analysis. **Figure 1** reports the PRISMA (2009) flow diagram of study selection, and specific search engine results are presented in *supplementary file 1*.

Data Extraction

Selected articles were subjected to three preliminary levels of screening: Title, Abstract and Full Text. The following data (when available) from articles was extracted using a pre-designed and piloted data-extraction pro-forma which included the following categories and sub-categories: (1) Referencing information, (2) Quality assessment, (3) Study characteristics (geographical origin, sample, objectives, design), (4) Type of measure, (5) Assessment to examine whether the properties of each measurement tool (construct validity, content validity, criterion validity, internal consistency, test-retest reliability, inter-rater reliability etc.) had been reported, (6) Main results/outcomes, (7) Additional further information relevant to the objectives. Results were recorded in an Excel spreadsheet and each study was assigned a reference identifier code (*supplementary file 2*).

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Study Assessment of Methodological Quality

Quality inclusion criteria were restricted by whether sufficient detail was provided to enable analysis of methodological quality and the evaluation of study measures. The five 'appraisal prompts for informing judgements about quality of papers' defined by Critical Interpretive Synthesis (CIS)[9] methodology were evaluated and tabulated for all included studies. CIS yielded quality appraisal nominal data scores for each question (1 yes, 0 no), with a cumulative score ranging from 0 to 5; scores ranged from 2 to 5. Ninety-four of the 105 studies achieved the maximum (5/5) cumulative rating score, six scoring (4/5), four scoring (3/5), and one scoring (2/5).

Data evaluation and synthesis

As the protocol anticipated, due to the heterogeneity of measurement tools in this review, and inconsistencies in methodology and reporting of results, a meta-analysis was not possible. Preliminary categorisation of the studies included data extraction of study characteristics, types of competence measures and measurement properties. A conceptual analytic synthesis of identified dimensions of similarity and difference in defining the construct of competence in this population was undertaken.

Patient and Public Involvement

Patients or the public were not directly involved in the design or planning of this study.

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RESULTS

The search strategy generated 105 articles selected for full data abstraction: 13 from PsychSOURCE, 19 MEDLINE/PubMed, 24 Cochrane Library – CENTRAL, 21 Web of Science, and 28 from specific paper retrieval forward citation searches at the journal site. Selected reviewed papers in the results section are referenced in the format[P:1,2,3,4,5], and individual measures are referred to as[M:1,2,3,4,5] (see *supplementary file 2* for assigned reference identifier codes and corresponding references).

Characteristics of Selected Studies

Included articles were published between 1990 and 2020. There was a progressive increase in the number of publications per year over this period (mean annual publications per year 0.3 from 1990 to 1999, 1.9 from 2000 to 2009, and 8.2 from 2010 to 2019), consistent with increasing research interest and societal awareness of the extent and consequences of medical error[P105]. The study designs included: two intervention, 22 measurement development, 26 measurement evaluation, 17 curriculum development, 40 clinical assessment, 21 training evaluation, and 11 experiential assessment; 26 studies comprised multiple categories.

Studies were restricted to English language, but without restriction on geographic origin and were derived from 18 countries. One study[P26] undertook data collection in two countries (USA and Canada), therefore data was counted in both. The majority of research was from the USA (n = 34 articles), UK (n = 17), Germany (n = 15), and Canada (n = 15). There were five studies from Australia[P3,35,64,80,102], four from the Netherlands[P23,49,58,69], and three from Korea[P13,61,73]. Two studies were from New Zealand[P5,12] and Denmark[P45,66]. Sweeden[P10], Lebanon[P36], Saudi Arabia[P46], Austria[P48], Belgium[P50], Norway[P59], Poland[P60], India[P79], and China[P103] were each represented by a single study. The percentage distribution of all articles geographic origin is presented in **figure 2**.

Competence Assessment Measures

From the 105 articles, 153 measures of competence were identified, 98 of which were unique to that study (a comprehensive list of all measures is presented in *supplementary data file*

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3). The competence assessment measures were categorised as either established 'in-practice measures' (IM) currently implemented for the assessment of doctors and medical students, or 'research-measures' (RM) developed or applied to assess competence for research evaluation, but not at the time of publication employed for occupational competence assessment. Thirty-eight studies examined IM, 45 examined RM, and 22 explored both RM and IM. Most (n = 76) of the selected studies were single-institution assessments - limiting the generalisability of findings, 25 were multicentric (based at multiple institutions), and four were not institution-based.

Measures were categorised by their procedural comparability, and seven dominant operational characteristics were identified: (1) Clinical Performance Assessment - was applied in 30.7% of measures (n = 38 IM, n = 9 RM), typically assessed using simulated patient ratings to assess clinical competencies such as communication skills and diagnostic reasoning in response to presented symptoms; (2) Anatomic Model Simulation – providing simulations of human anatomy in order to test knowledge and used in 6.5% of measures (n = 9 IM, n = 1 RM); (3) Psychometric Assessment – predominantly evaluated personality, aptitude, reasoning and emotionality, and conducted in 7.8% of measures (n = 2 IM, n = 10 RM); (4) Patient Records - to assess the medical record keeping accuracy and completeness, were utilised in 1.3% (n = 2 IM); (5) Computerized Testing – included a variety of computerised assessments, used in 9.8% of measures (n = 1 IM, n = 14 RM); and (6) Written Testing – included multiple-choice and short-answer questions in traditional pen-and-paper testing in 31.4% of measures (n = 19 IM, n = 29 RM). The prevalence of each measurement category demonstrated a preference for the application of written testing and clinical performance assessment in the assessment of competence. Fifty-six (36.6%) measures applied a unidimensional procedure to assess the target domains, and 97 (63.4%) operated multidimensional procedures.

Established In-practice Measures

Competence was assessed through a variety of IM methods. Objective structured clinical examinations (OSCEs) typically involve patients or simulated patients and the most duplicated measure (in 20 articles). OSCEs are frequently considered[P:53] the 'gold standard' for assessing competence skills, as they are high-fidelity, can be reliable, and are regarded as valid because of

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their ability to differentiate between occupational levels of aptitude and specialisms. The use of standardised patients (SPs) endeavours to replicate doctor-patient consultations can identify variation in clinical practice between doctors in a realistic simulated setting. SP single-case assessments were used to measure specific competencies in nine of the studies.

Anatomic model simulation testing, employed in 10 measures, is widely used as a safe method to develop procedural competence in early medical education. However, whether the anatomic-model patients are the most valid means of assessing procedural competence is widely debated[P38]. Workplace-based assessment (WBA) such as mini-clinical evaluations (Mini-CEX) and direct observation of procedural skills (DOPS) are heavily integrated in medical education. However, the Mini-CEX and DOPS lack standardisation, cannot be used with large numbers of participants simultaneously, are costly, time consuming, and transferability of skills is not always guaranteed in a simulated assessment setting; particularly in a workplace with high levels of social, psychological and physiological stress[P:20]. Four studies[P:20,29,79,89] examined 360-degree assessments (i.e., rating evaluations from colleagues within the workplace)

Written tests provide reliable assessment of several domains, particularly knowledge and psychometric assessment. Three studies[P:2,95,97] used the Jefferson Scale of Physician Empathy (JSE), one [P:3] employed the Jefferson Scale of Physician Empathy – Student Version (JSPF-S). Written tests measuring knowledge included two studies using the US Medical Licensing Examinations (USMLE), and three studies that included the Medical College Assessment Test (MCAT) - an assessment used by many medical schools as part of the entry-selection process. However, written tests are considered insufficient to measure behavioural clinical competence [P:20]; therefore, there has been a traditional reliance on the judgement of competence by educators or seniors in demonstration-observation tests of clinical performance skills. The combination of a knowledge test (written assessment) with an applied test of clinical skill (OSCE) was shown to increase predictive validity versus either alone[P:12]. Vignettes (five studies) and written case simulations have been widely used to assess competence in both clinical training and experimental research, because of their ease of administration, negligible cost, and objective quantifiable results. However, they do not assess important social interaction skills such as history taking, nor assess

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competence in a real-world situation including occupational stressors. Amongst IM measures there was significant variability in testing procedures, content domains and inconsistent assessment outcomes, therefore these were not grouped for evaluation purposes.

Contemporary Experimental Research Measures

Electronic-technology has an increasing role in healthcare, but raises new challenges for competence assessment; one study presented an electronic platform[M:149] that automatically assesses a doctor's competence from online textual consultations with patients utilizing a novel machine that provides auto-evaluation based on patient satisfaction following online consultation. In addition to patient evaluations, innovative simulation-based medical education (SBME) and assessment provides alternatives to practice and assessment without risk to patients[P:26].

In an era where doctors' retirement age has extended, further cognitive competence considerations have emerged. Forty-two measures in 37 of the studies assessed cognitive competence, publication dates indicating that research interest in this area has markedly risen in recent years (7.6% of the studies between 2000 and 2009, compared to 26.7% from 2010 to 2020). Neuropsychological screening found[P:81] that many of the doctors who fell significantly below expected competence assessment levels had cognitive impairment sufficient to explain their diminished competence. Another study developed 'the Mini-NeuroCart'[M:80], a cognitive and psychomotor test battery for assessment of subjective and objective measures of alertness, mood, concentration and self-assessed procedural competence, and showed that the competence of postcall surgeons was similar to, or worse than that of ethanol-intoxicated surgeons.

Competence Domains

Studies were categorised according to the competence assessment domains. The list of measures in each domain is presented in *supplemental material file 4*. Twelve ascendant domain categories based upon their prevalence in the selected measures were identified, with descriptors of these in **table 1**.

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Table 1 Domain Descriptors in Descendent Order of Prevalence in All Measures

Number	Domain	Prevalence	Descriptor
1	KNOWLEDGE	55.5%	The acquisition and retention of learning.
2	PROCEDURAL COMPETENCE	45.1%	The ability to perform specific technical skills/procedures required for the occupational obligations.
3	JUDGEMENT & BIAS	41.8%	An ability to nullify or negate judgement biases to competently work in the benefit of patients and colleagues.
4	COMMUNICATION	40.5%	The ability to engage patients and colleagues in effective interaction.
5	PROFESSIONALISM	29.4%	To uphold the expected traditional occupational behavioural standards of the profession, and regulatory code of conduct.
6	COGNITIVE COMPETENCE	27.4%	To effectively utilize the cognitive resources required for the selection and application of skills.
7	ENVIRONMENTAL COMPETENCE	23.5%	This domain relates to the ability to maintain acuity in environmental adversity.
8	COPING COMPETENCIES	21.5%	Competence to psychologically manage the complex occupational exigencies.
9	SELF-ASSESSMENT ACCURACY	20.9%	Self-assessment and accurate self efficacy in continuous self-monitoring of competence.
10	EMPATHIC COMPETENCE	20.2%	Empathic Competence describes the awareness and responsiveness to patients' and colleagues' feelings and experiences.
11	ETHICS & PATIENT SAFETY	16.9%	A commitment to exemplar medical practice by prioritising patient safety and ethical standards.
12	PHYSIOLOGICAL COMPETENCE	6.5%	Physiological competence describes a state of optimal functioning that supports the physical and psychological workplace demands.

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Consideration of Each Domain Within the Literature

Knowledge (**Domain 1**), deemed as the primary proximal determinant of competence[P:11] was the most commonly assessed competency in IM, historically assessed through written examinations[P:31], contemporary measures endeavour to combine knowledge with procedural assessment. Concordantly, only 27.2% of measures conformed to traditional written-test approaches, though 6.4% of computerised test measures were a comparable format.

Within **Domain 2** (Procedural Competence), 54.6% of the measures employed a clinical performance test; other measures included patient vignettes, written or computerised tests (30.2%), and anatomic model simulation (15.6%).

Domain 3 (Judgement and Bias) was addressed in 41.8% of measures and incorporated competence with diversity, and awareness of bias in an occupational context. Cultural competence was found to be lacking in one study[P:8] as a result of an interaction between race and treatment outcomes as there was a higher likelihood of doctors escalating the use of opioids for African American patients. Sexual and gender minority individuals experience high rates of discrimination when seeking healthcare, contributing to patient care disparities[P:68]. Furthermore, doctors reported discomfort when treating people who have disabilities[P:96], as assessed by the Scale of Attitudes Toward Disabled Persons (SADP)[M141].

Domain 4 (Communication): Competent doctor-patient and inter-professional communication is included within this domain. Ineffective communication with colleagues has been associated with reduced patient safety and risk to care quality[P:71], and the most frequent cause of complaints against doctors is related to poor communication[P:42].

Though doctors are expected to uphold the customary standards of professionalism (**Domain 5**), there was no consensus on its definition[P:77], and an absence of explicit standardised factors measured within this domain[P:78]. Most (73.8%) of the measures used subjective rating scales; many within OSCE and WBA. Remarkably, although 36 IM measured professionalism, only 9 RM assessed this domain.

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Cognitive competence (**Domain 6**) denotes the specific cognitive resources required for the selection and application of skills such as diagnostic and clinical reasoning. Wechsler Adult Intelligence Scale (WAIS-IV) assessments have found cognitive resource differences between doctors who experience difficulties, and those who do not[P:70]. The cognitive concept of 'mental workload' perceives the brain having a limited capacity to process stimuli, research has found that high levels of workload are associated with error and poor competence[P:19]. Traditional measurement of task performance does not necessarily predict competence when a combination of tasks must be undertaken under time pressure resulting in cognitive overload, and the 'task-shedding' of key competencies resulting in risk of impaired competence.

Environmental Competence (**Domain 7**) employs measures that account for (or simulate) the context of occupational pressures in assimilating and responding to environmental context[P:72] using real-time information management[P:54]. One study[P:19] examined the effect of multitasking on competence in medical students using a secondary task method employing a venepuncture procedure[M:32] alongside an electronic application[M:31] which demanded a tapping response when the device vibrated, mental workload was measured in time delay and effects were observed on communication deterioration. Measures that included time (chronometry)[P:15,19,26,34,46] highlighted that timed test increased the reliability and validity of measuring diagnostic accuracy[P:15], as time-pressure has been found to disrupt the dynamic interaction between reasoning systems influencing diagnostic tasks[P46]. Conclusions about competence may be flawed unless contextual environmental factors are considered[P:72].

Doctors are expected to handle limited resources, pressure, and life or death performance outcomes that often carry an emotional load; 'Coping Competence' (**Domain 8**) is therefore paramount[P:72]. Distress and inability to cope has been associated with medical errors[P:105]. Therefore, the ability to cope with environmental, cognitive, physical and emotional workplace demands is key to competence and occupational fulfillment[P:104].

The ability to accurately 'self-assess' and work at a level of occupational aptitude. (**Domain 9**) is key to the identification of difficulties, and safe practice. RM focussed more heavily on self report measures (n = 27) than IM (n = 5). Concerningly, one study[P:3] found that self-rated empathy and observer ratings were not associated, and another[P:7] found no

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correlation between confidence scores and clinical or knowledge abilities. A doctor's ability at any career stage to accurately self-assess their competence acquisition provides assurance of working within scope of practice[P:14].

Medical educators and professional regulators recognise the importance of Empathic Competence (**Domain 10**); strong associations were found between observer ratings of empathy and OSCE scores[P:3]. The prominence of empathic competence within the selected measures was comparable in RM (n = 16) and IM (n = 15). In medical education research[P:95], empathic competence has been demonstrated to increase in male trainees and decrease in females during the course of training. Importantly, empathic competence has been found to be strongly associated with patient satisfaction[P:97].

Domain 11 relates to adherence to ethical standards and commitment to patient safety. This domain was addressed in regulatory assessments[e.g.M:129,130,138], rating scales assessed by senior doctors[e.g.M:1,9,91,117,126], WBA's[e.g.M:146,147], patient evaluations[M:36], and values-based measures such as a questionnaire survey about notions of a 'good doctor'[M:8]. Neuropsychological testing[M:123] has been found to be a strong predictor of risk to competence[P:86]; a consistent decline in competence has also been identified in recertification exams in America related to years since graduation[P:82]. Another study[P:75] developed and tested an OSCE designed to directly assess socio-cultural dimensions of patient safety competency[M:117], demonstrating sufficiently reliable station scores in this domain. Engagement with continuous competence improvement and collaboration with colleagues (rather than isolated practice) has been identified as a protective factor[P:28].

Physiological competence (**Domain 12**) emerged as a domain rarely considered in assessment. RM measures within this domain included psychometric measures of burnout[M:18,52], the Epworth Sleepiness Scale[M:151], and the Medical Outcomes Study Short Form (SF8) psychological and physical health survey[M:48]. Traditional training and assessment typically occurs in optimum daytime working conditions, which arguably[P:44] neglects nocturnal circadian effects. One study[P:44] found that training delivered during a simulated night shift in an emergency department was effective in significantly increasing medical students' self-efficacy. Another study[P:48] used electroencephalographic recordings and measured the

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competence of doctors in the course of 24-hr shifts with and without afternoon rest, demonstrating nocturnal cognitive deactivation without rest, and a stimulating vigilance promoting effect of the resting period. Only 2/10 psychophysiological measures[M:129,130] were IM, both were regulatory assessments.

Figure 3 reports the distribution of domains of competence assessed in selected studies reporting in-practice assessments of competence (% of IM), and those from research studies (% of RM). There was notably some variation between the IM and RM in domain prevalence. There was less focus on the 'knowledge' domain in the RM studies. However, the largest discrepancy between the RM and the IM papers was for the 'professionalism' domain, which was rarely included (n = 9) in RM compared to IM (n = 36) ($\chi^2 = 26.6$; $p < 0.001$).

Figure 4 presents the percentage of the procedures that were used in the measures within each of the twelve domains. Therefore, the outcome of the construct domain consolidation of measures of competence was the identification of 12 domains. There were significant differences in the predominant domains assessed in IM versus RM studies, with less focus on professionalism and knowledge-based domains in the latter.

Instrument Properties

Instrument properties were evaluated for their attention to the reporting of measurement properties such as reliability, validity, and interpretability. Despite the preponderance of identified measures, there is a dearth of evidence for their reliability and validity; 53.5% of the identified measures provided reliability and validity data (37/73 IM and 45/80 RM), whereas 21.5% of studies described measures without reference to either their reliability or validity (16/73 IM and 17/80 RM).

Measurement content validity was predominantly based upon citation in several of the measures[e.g.M:46,47,48,50,52,64,65,114] rather than validation against an agreed standard measure. With regard to internal consistency, a Cronbach's alpha (α) value was reported for 39 reviewed measures. Cross-cultural validity was addressed in 29.4% of measures. Predictive validity was addressed by a small number of studies to measure the predictive value of pre-medical school

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assessment on later performance, and the predictive value of cognitive assessment in predicting subsequent impaired competence in aging doctors[P:81,M:123]. Computerised, written, and oral measures were typically single-domain measures of knowledge but reported higher levels of validity.

Details on measurement interpretability were found for 119 reviewed measures. As many established measurement methods were institution-specific there was substantial variability in testing procedures, content domains and subsequent outcomes. However, practice and test-retest effects in longitudinal studies received minimal consideration[P:42, 81]. Testing measures of competence in real practice using double-blind observations are evidently unfeasible for logistical and ethical reasons, however workplace-based 360-degree feedback from a doctor or trainees colleagues provides an alternative whereby varied sources of in-practice observations with real patients and there is evidence for the validity of this method[M:33,43].

Measurement Assessment Context

Increasing effort has been dedicated to enhancing the realism of technical skills training by introducing SP's and role-plays[P:51]; 45.4% of the reported measures (53/73 IM and 30/80 RM) were designed to be assessed in a simulated context, and just 20.9% with real patients (only 18 of which were IM), 33.7% of measures used neither. The majority assessed competence in either a workplace-based assessment (n = 19), or (more commonly) a simulated assessment (n = 67). However, the acceptability and feasibility of real patient measures rather than simulated context assessment encounters has significant cost, time and ethical barriers. However, there has been increased emphasis in more recent studies on assessments which attempt to mimic real world clinical situations.

Positional Orientations

Studies assessed competence measurement from two positional orientations; assessor and assessee. Measurement of assessee competence was categorised by four populations; medical school applicants (n = 3), medical students (n = 60), junior doctors (n = 19) and fully trained (senior) doctors (n = 20). The assessment of medical school applicants for core

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competencies received minimal attention (3 studies) in the selected literature; one research study[P:9] utilized a video-based situational judgement test (SJT)[M:16] assessing social competencies, the second[P:58] addressed an assessment framework measure[M:91], and the third[P:60] employed a coping responses inventory[M:94] in addition to a state entrance examination[M:95]. Medical students were the target population in the majority of studies, as a competency-based curriculum is becoming a dominant organizing framework for medical education and assessment[P:47], and its emphasis was reflected in its research dominance in the literature. Only three studies examined multiple assessee populations; one[P:60] studied medical school applicants, medical students, junior doctors, and fully qualified doctors using psychometric burnout[M:94] and knowledge test[M:95] measures, and found that the assessment of coping competence at medical school enrolment was a predictor of longitudinal professional competence and career development. The second[P:72] (medical students, junior doctors, and fully qualified doctors) used a group evaluation[M:113] to explore the contextual-environmental factors that affect competence assessment, and the third study[P:92] found that vignettes[M:135], SP's[M:136], and chart abstraction[M:137] measures were a valid and comprehensive method to measure competence in junior doctors and fully qualified doctors.

Assessor perspectives are important in the interpretation of competence outcomes. Assessor-performed analogue rating scales were used in 47 measures (31 IM and 16 RM). Evaluative standpoints of assessors varied between the selected studies and were described as follows: Fifty measures (54.8% of IM, 12.5% of RM) employed supervisory occupational appraisal of a junior or student. Six measures (4.1% of IM, 3.8% of RM) used peer assessment. Twenty-two measures (19.2% of IM, 10% of RM) utilised patients or SP's who can provide a key role in evaluating competence as judged by their direct experience of the doctor or trainee. Four measures (2.7% of IM, 2.5% of RM) used disciplinary/regulatory assessment procedures designed to assess a comprehensive range of domains. Finally, 37 measures (8.2% of IM, 38.8% of RM) involved self-assessment of competence. Therefore, a diverse range of assessor standpoints influence competence assessment and providing feedback is arguably[P:23] a complex affective process and the assessment is determined by the assessor's cognition, beliefs, and emotions. Inter-rater reliability is pertinent to the interpretability of selected measures in this review as the majority of multi-domain IM evaluations of clinical performance, such as OSCE's, and the mini-CEX rely extensively on rater-assessment. Inter-rater reliability was reported for a small number of studies[e.g.P:51,67,75,77,89], however many were monocentric studies.

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DISCUSSION

This systematic review sought to identify and evaluate measures of competence in doctors, medical students and those applying for entry to medical school. 105 articles were identified that included a broad range of 153 competence assessment measures. The principal findings were as follows: Very few studies conceptualised competence through definition or operationalising the construct. Therefore, although competence is considered paramount for safe practice, no consensus in the selected literature exists regarding its conceptualisation, nor is there measurement standardisation. Twelve domains used to conceptualise and measure competence were identified. The deconstruction of each domain and its relevance yielded strong operational purpose in 11 of the domains but identified inconsistencies and procedural subjectivities regarding Professionalism (Domain 5). Assessment of professionalism in the studies lacked reliable measurement evaluation tools and was open to implicit judgment biases; this review therefore questions whether further objective assessment of professionalism may be identified and validated. In contrast, judgement and bias (Domain 3) included a range of specific measures applied to assess competence with diversity and judgement orientations; the findings that patient characteristics were commonly associated with treatment outcomes requires continuous examination from a training and assessment perspective to ensure non-discriminatory ethical treatment of all patients[10]. However, it is important to acknowledge that the competence domain categories are not necessarily indicative or proportionally reflective of all globally accepted assessment tools. There are likely varying conceptualisations of competence in different cultures, and in different clinical, social and economic contexts, therefore further research on geographic and cultural construct analysis is advocated to explore potential differences in worldwide assessments.

Interestingly, the distribution of competence domains differed between IM and RM; the RM reflecting more explicit and focused consideration of the contribution of behaviouristic (skills and performance), systemic pressures, psychophysiological (cognitive, emotional, health), and vantage-point (beliefs, perceptions and attitudes) factors as conceptually relevant to competence. A further follow-up exploration to examine which of the RM are being implemented in systemic operational practice would be worthwhile. It is noteworthy that findings related to the validation of the measures were often limited or entirely absent, therefore further validation of existing measures is recommended. There is a broadly held assumption that assessment formats with singular competence domains are limited[11] whereas utilising

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comprehensive multidimensional assessment measures will minimise deficiencies[12]. Furthermore, the majority (n = 76) of selected studies were single-centre assessments. To address this, we encourage multicentric testing of competence assessment to offer greater generalisability of the data. Correspondingly, the sharing of assessment measures between institutions may encourage greater homogeneity in practices.

The assessment of some domains such as coping competence are considered by many medical schools to be crucial in identifying those with the necessary attributes to train to be a doctor. These domains may therefore be considered as part of the continuum of assessment of competence throughout a medical career. These were therefore included in this systematic review. This continuum approach is consistent with the increasing recent interest in programmatic longitudinal assessment of both students and doctors using multiple assessment techniques, in which individual assessment elements are part of a process of gathering information contributing to the overall assessment rather than being used on their own to make decisions. In this model these assessments contribute to the learning process, breaking down the traditional dichotomy between formative and summative assessment [13]. The findings of this systematic review support this, because it is clear that individual assessments taken in isolation each have their limitations. Furthermore, the inclusion of both formative and summative components in this longitudinal assessment enhances the learning process.

A wide range of global competence measures were reported in the selected studies. These included written assessments, OSCEs, anatomic simulations, and workplace-based assessments. The combination of a written test with a clinical skill OSCE test has been demonstrated to increase the predictive validity of either alone. In research studies assessing competence there is considerable interest in simulation-based assessments and some have focussed on cognitive assessment which may be of particular importance with many doctors working beyond retirement.

Many assessments of competence are contingent upon assessor subjectivity. Whilst some studies demonstrated attempts to assess inter-rater reliability, this is an important issue that requires further attention. Assessor bias is therefore a worthwhile area of further research due to concerning discrepancies in assessment accuracy that have been attributed to unreliable self-assessment,[14] inconsistent applications of organizational standards resulting in a lack of reliability, and unclear rating scale criteria. Assessments involving direct observations

inherently include automatic judgement processes, and don't necessarily provide objective assessment,[15]. However, it has been argued,[16] that standardisation may be a constraint on authentic assessment interpretations and delineations. Approaches using 360-degree global rating evaluations by multidisciplinary multilevel colleagues who interact with a doctor or trainee routinely may improve objectivity through a comprehensive range of perspectives compared to single-assessor perspectives,[17]. This review highlighted the inconsistent judgement acuity of assessors as paramount to competence assessment[e.g.P:1,5,7,17,18,20,28,36,47,52,69,101]. When evaluating instrument properties, the consideration of feasibility is paramount, as all measures require resources for systemic implementation. Furthermore, the number of individuals being assessed, the time constraints, financial resources, the interpretability, the validity and reliability of measures are key considerations. When reviewing these points, many experimental measures are unfeasible as routine in-practice measures due to resource availability and/or ethical and time constraints.

There was unanimity in the literature that competence assessment throughout a doctor's career is necessary to protect patients and the profession,[1]. Therefore, continued exploration of evidence-based measures to assess competence throughout doctors' training and career is advocated. This review highlights the absence of consensus on the domains relevant to competence, and a lack of consistency in existing clinical measurement standards. In conclusion, the findings of this study strongly support the use of multidomain evidence-based measurement-selection, and attending to developing measures that replicate the unique systemic context through the evaluation of all domains relevant to competence is recommended as a continued focus of research in this area.

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FIGURE LEGENDS

Figure 1 reports the flow diagram of the review search strategy and study selection based on PRISMA (2009) guidelines

Figure 2 presents the percentage distribution of all articles' geographic origin

Figure 3 reports the distribution of domains of competence assessed in selected studies reporting in-practice assessments of competence, and those from research studies

Figure 4 presents donut charts representing the percentage of the procedures that were used in the measures within each of the twelve domains

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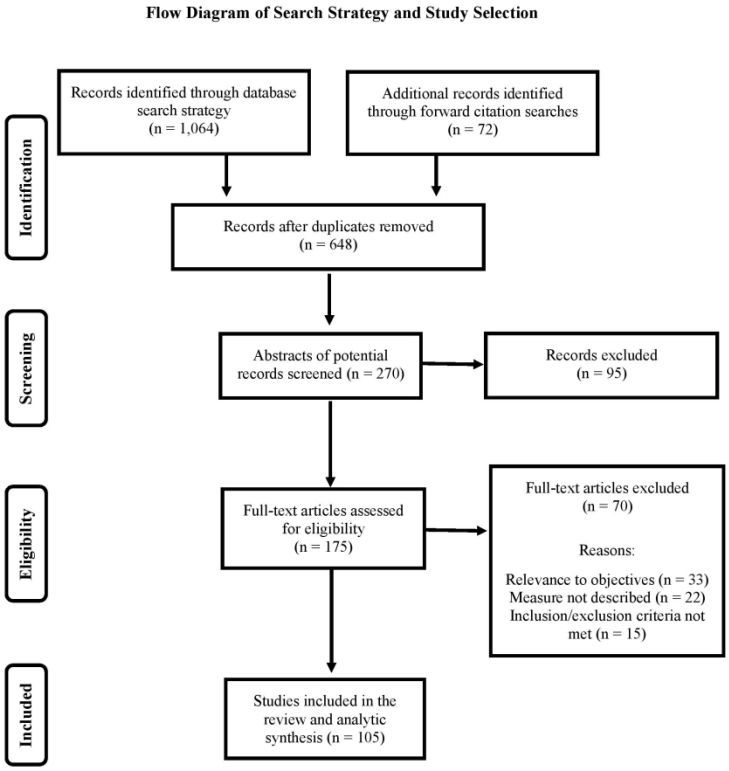


Figure 1 Flow Diagram of Search Strategy and Study Selection Based on PRISMA (2009) Guidelines

Figure 1 reports the flow diagram of the review search strategy and study selection based on PRISMA (2009) guidelines

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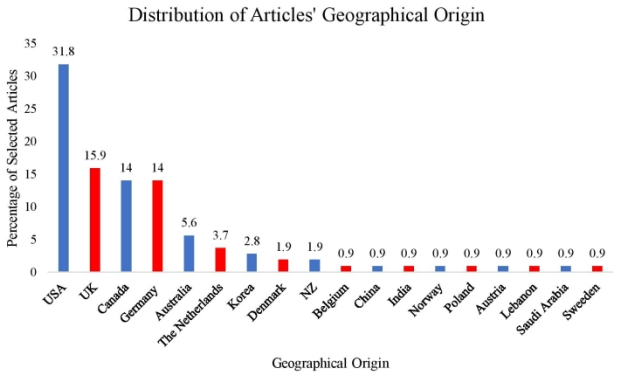


Figure 2 The Percentage Distribution of Selected Articles' Geographical Origin

Figure 2 presents the percentage distribution of all articles' geographic origin.

209x297mm (300 x 300 DPI)

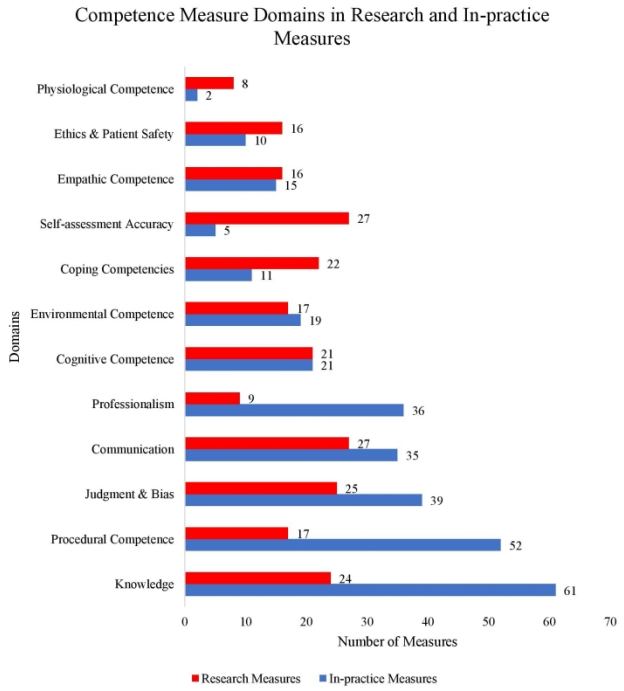


Figure 3 Competence Domains in Research and In-practice Measures

Figure 3 reports the distribution of domains of competence assessed in selected studies reporting in-practice assessments of competence, and those from research studies

209x297mm (300 x 300 DPI)



Figure 4 Donut charts representing the distribution of procedural variation within each competence domain

Figure 4 presents donut charts representing the percentage of the procedures that were used in the measures within each of the twelve domains

209x297mm (300 x 300 DPI)

MEASURING COMPETENCE IN DOCTORS

Supplemental Material File 1

Study Search and Selection Results

Electronic Search Engine		Total Identified Records	Number Following Deduplication	Relevant Titles Selected for Abstract Screening	Full Text Articles Assessed for Eligibility	Studies Included in Analytic Synthesis
PsychSource	English Language Search	221	221	31	17	7
	American-English Search	224	42	12	9	6
Medline/ PubMed	English Language Search	132	65	35	19	10
	American-English Search	179	57	39	14	9
Cochrane Library – Central	English Language Search	53	41	20	19	13
	American-English Search	117	56	17	18	11
Web of Science	English Language Search	58	52	19	14	10
	American-English Search	80	42	25	15	11
Specific Paper Retrieval at Journal Site – Forward Citation Search		72	72	72	51	28
Total		1, 136	648	270	175	105

Note. Search results are reported in the sequence they were undertaken

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Supplemental Material File 2

Assigned Reference Identifier Codes and Full References

Identifier	Reference	Identifier	Reference
P1	Boursicot (2006)	P46	ALQahtani *et al.*, (2018)
P2	Hojat *et al.*, (2002)	P47	Yeates *et al.*, (2012)
P3	Ogle *et al.*, (2013)	P48	Frey *et al.*, (2002)
P4	Maudsley *et al.*, (2007)	P49	Huizinga *et al.*, (2018)
P5	Wilkinson & Frampton (2003)	P50	Aper *et al.*, (2012)
P6	Williams & Klamen (2015)	P51	Nikendei *et al.*, (2007)
P7	Morgan & Cleave-Hogg (2002)	P52	Hall *et al.*, (2015)
P8	Burgess *et al.*, (2011)	P53	Truncali *et al.*, (2011)
P9	Frölich *et al.*, (2017)	P54	Chumley *et al.*, (2006)
P10	Dahlin *et al.*, (2010)	P55	Ludwig *et al.*, (2018)
P11	Kell *et al.*, (2014)	P56	Windish *et al.*, (2005)
P12	Wilkinson & Frampton (2004)	P57	Bugaj *et al.*, (2018)
P13	Oh *et al.*, (2016)	P58	Schreurs *et al.*, (2018)
P14	Homberg *et al.*, (2019)	P59	Gude *et al.*, (2017)
P15	Monteiro *et al.*, (2019)	P60	Tartas *et al.*, (2016)
P16	Radabaugh *et al.*, (2019)	P61	Han & Chung (2016)
P17	Harendza *et al.*, (2019)	P62	Devine *et al.*, (2015)
P18	Madrazo *et al.*, (2018)	P63	Roland *et al.*, (2015)
P19	Woods *et al.*, (2018)	P64	Carr *et al.*, (2014)
P20	Prediger *et al.*, (2019)	P65	Mehdizadeh *et al.*, (2014)
P21	Gerard *et al.*, (2018)	P66	Ammentorp *et al.*, (2013)
P22	Renaud & Côté (2017)	P67	Terregino *et al.*, (2019)
P23	Sehlbach (2020)	P68	Zelin *et al.*, (2018)
P24	Olsen *et al.*, (2019)	P69	Pool *et al.*, (2018)
P25	Burnett *et al.*, (2019)	P70	Patterson *et al.*, (2017)
P26	Petrosoniak *et al.*, (2019)	P71	Zabar *et al.*, (2016)
P27	Siau *et al.*, (2019)	P72	Stroud *et al.* (2015)
P28	Kain *et al.*, (2019)	P73	Park *et al.*, (2015)
P29	Prediger *et al.*, (2020)	P74	Dong *et al.*, (2015)
P30	Giesler *et al.*, (2011)	P75	Ginsburg *et al.*, (2015)
P31	Bodamer *et al.*, (2015)	P76	Ferenchick & Solomon (2013)
P32	Beckman *et al.*, (2012)	P77	Thomas *et al.*, (2011)
P33	Hautz *et al.*, (2017)	P78	Ginsburg *et al.*, (2010)
P34	Krautter *et al.*, (2011)	P79	Jani *et al.*, (2017)
P35	Liddell *et al.*, (2002)	P80	Spittal *et al.*, (2015)
P36	Yazbeck-Karam *et al.*, (2014)	P81	Turnbull *et al.*, (2006)
P37	Engerer *et al.*, (2016)	P82	Caulford *et al.*, (1994)
P38	Posner *et al.*, (2013)	P83	Grace *et al.*, (2014)
P39	Mc Evoy *et al.*, (2016)	P84	Lehmann *et al.*, (1990)
P40	Snow *et al.*, (2016)	P85	Meagher *et al.*, (2009)
P41	Hoefer *et al.*, (2017)	P86	Mehdizadeh *et al.*, (2014)
P42	Shapiro *et al.*, (2009)	P87	Unwin *et al.*, (2019)
P43	Krautter *et al.*, (2014)	P88	Devitt *et al.*, (1998)
P44	Stroben *et al.*, (2016)	P89	Joshi *et al.*, (2004)
P45	Christiansen *et al.*, (2014)	P90	Smith *et al.*, (2019)

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Identifier	Reference	Identifier	Reference
P91	Santen *et al.*, (2013)	P99	Green *et al.*, (2017)
P92	Peabody *et al.*, (2000)	P100	Mitchell *et al.*, (2011)
P93	Humphrey *et al.*, (2011)	P101	Tham *et al.*, (2017)
P94	Gollehon *et al.*, (2017)	P102	Shulruf *et al.*, (2019)
P95	Austin *et al.*, (2007)	P103	Shi *et al.*, (2018)
P96	Kirby *et al.*, (2011)	P104	Trockel *et al.*, (2018)
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MEASURING COMPETENCE IN DOCTORS

Supplemental Material File 3

Comprehensive List of All Identified Measures

Reference Identifier	IM/ RM	Outcome Measure	Measure Identifier
P1	IM (A)	The Angoff Method (OSCE rating scale)	M1
P2	IM	US Medical Licensing Examinations (USMLE)	M2
P2	IM	Medical College Admission Test (MCAT)	M3
P2	RM	Jefferson Scale of Physician Empathy	M4
P3	RM	Jefferson Scale of Physician Empathy – Student Version (JSPF-S)	M5
P3	IM	Objective structured clinical examination (OSCE)	M6
P3	IM	Rating Scales for the Assessment of Empathic Communication in Medical Interviews (REM)	M7
P4	RM	Questionnaire survey about notions of a 'good doctor' 6 brief questionnaire surveys	M8
P5	IM (A)	Performance Rating Instrument - A Trainee Intern Rating Form	M9
P6	RM	Twenty Questions (TQ) game performance	M10
P6	IM	Medical College Admission Test (MCAT) Biological Science Test Score	M11
P7	RM	25-point prior experience and confidence questionnaire	M12
P7	IM	A pre-determined critical event simulation	M13
P8	RM	Vignettes on chronic pain (assessed by treatment outcome)	M14
P8	IM	The short Medical Authoritarianism Scale (MA-S)	M15
P9	RM	A video-based situational judgment test (SJT) assessing social competencies	M16
P10	RM	Performance-based self-esteem (PBSE)	M17
P10	RM	Oldenburg Burnout Inventory (OLBI)	M18
P11	IM	Standardised Patient Examination (SPE)	M19
P11	IM	Opinions about Physicians Interactions with Patients (OPIP)	M20
P11	IM	Medical College Admission Test (MCAT) GPA	M21
P12	IM	OSCE combined with a written examination	M22
P13	RM	Anderson's Epistemological Beliefs Questionnaire (EBQ)	M23
P13	IM	SP: Korean Clinical Performance Examination (CPX)	M24
P14	IM (A)	The CanMEDS Physician Competency Framework	M25
P15	IM	Written Vignettes	M26
P15	RM	Time-pressured Vignette Assessment	M27
P16	IM	US Medical Licensing Examinations (USMLE)	M28
P17	RM	Group Assessment Performance (GAP) Test	M29
P18	IM	OSCE self-assessment measure on a 6-point Global Rating Scale (GRS)	M30

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P19	RM	Workload InsightTM Application (Chronometry measure of task)	M31
P19	IM	The Secondary Task Method – Venipuncture (Simulated task performance rated using a 16-point checklist)	M32
P20	RM (A)	A 360-degree Assessment of Competencies	M33
P21	RM	PediatricSim Electronic Game	M34
P22	RM	The 5-item Doctor-Patient Communication Competency Instrument for Medical Students (SP)	M35
P23	RM	Patient Evaluation of Physician Competence	M36
P24	RM	Diagnostic Competency Framework	M37
P25	IM	Simulated Patient (SP) with simulated crisis parameters of mortality and morbidity	M38
P26	IM	7-point Global Rating Scale (GRS)	M39
P26	RM	Chronometry	M40
P27	IM	Direct Observation of Procedural Skills (DOPS)	M41
P28	RM (A)	Qualitative Experiential Knowledge	M42
P29	RM	360-degree Assessment	M43
P30	RM	The Freiberg Questionnaire to Assess Competencies in Medicine	M44
P31	RM	Internal Medicine Standardized Simulation Based Examination (SSBE)	M45
P32	RM	The Interpersonal Reactivity Index (IRI)	M46
P32	RM	The Maslach Burnout Inventory	M47
P32	RM	Medical Outcomes Study Short Form (SF-8)	M48
P32	IM	The Mini-Clinical Evaluation Exercise (mini-CEX)	M49
P32	IM	In-training Examinations (ITE)	M50
P33	IM	Global Rating Scale (GRS)	M51
P33	RM	The Test of Self-Conscious Affect, Version 3 (TOSCA-3)	M52
P34	RM	Chronometry	M53
P34	RM	Objective Performance Ratings by Independent Assessors of Videographed Procedure	M54
P35	IM	Procedural Assessment of Clinical Skill (Injection & Suturing)	M55
P35	RM	Procedural Skills Confidence Survey	M56
P36	IM	OSCE (SP)	M57
P37	IM	Student Self-Ratings of Communication Competence (VAS)	M58
P38	IM	SP-PTT (part-task trainer) hybrid: OSCE	M59
P38	IM	PTT (part-task trainer): OSCE	M60
P39	RM	Written reflective assignment	M61
P39	RM	Rating Scale - self assessment	M62
P40	IM	OSCE	M63
P40	IM	Doctors' Interpersonal Skills Questionnaire (DISQ)	M64
P41	IM	OSCE Station (videotaped)	M65
P42	RM	Self-Assessment of Interpersonal Competence Questionnaire (SAICQ)	M66
P42	RM	The Staff-Patient Interaction Scale (SPIR)	M67
P42	RM	Interpersonal Skills Rating Scale (ISRS) rated by SP's	M68

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P43	RM	Self efficacy rating (Likert scale)	M69
P43	IM	SP clinical performance assessment by assessor direct observation (DO)	M70
P44	RM	Self-reported preparedness (self efficacy) Likert questionnaire	M71
P44	IM	SP observation using the Calgary-Cambridge Observation Guide (CCOG)	M72
P45	IM	Radiological (X-ray) diagnostic accuracy	M73
P45	RM	Diagnostic Confidence Rating (Likert scale)	M74
P46	RM	Diagnosis using written clinical vignettes on Qualtrics	M75
P46	RM	Chronometry	M76
P47	IM (A)	mini-CEX	M77
P48	RM	Computer-analyzed Electroencephalography	M78
P48	RM	Pauli Test	M79
P49	RM	The Mini-NeuroCart	M80
P50	RM	Self efficacy scale	M81
P50	RM	Responses to a standardised video case	M82
P51	IM	OSCE test - Global rating scale of communication skills - binary checklist	M83
P52	RM	Analogue Patient Satisfaction Questionnaire	M84
P53	IM	OSCE SP checklist	M85
P54	IM	SP case	M86
P55	IM	Formative exit exam (medical school module) on clinical reasoning and diagnostic accuracy	M87
P56	IM	SP's who rated students on a 5-point scale for communication skills	M88
P56	IM	Diagnostic Thinking Inventory	M89
P57	IM	Procedural Protocol Instrument (IPPI) Global rating scale - blind video assessed	M90
P58	RM	CanMEDS Assessment Framework Measure	M91
P59	RM	The Oslo Inventory of Self-Reported Communication Skills (OSISCS)	M92
P59	IM	SP - The Arizona Communication Interview Rating Scale (ACIR)	M93
P60	RM	Coping Responses Inventory (Moos)	M94
P60	IM	State Examination for Medical Doctors	M95
P60	RM	Maslach Burnout Inventory (MBI)	M96
P61	IM	CPX - Clinical Performance Examination (SP)	M97
P61	IM	OSCE	M98
P62	IM	MRCP (UK)	M99
P62	IM	MRCGP AKT (Applied Knowledge Test)	M100
P63	RM	Self-evaluation meta-planning focus group	M101
P64	IM	The Junior Doctor Assessment Tool (JDAT)	M102
P65	IM	GMC test of Competence (ToC)	M103
P65	IM	OSCE	M104
P65	RM	Self-Assessment Questionnaire - ranking in comparison to other doctors	M105
P66	IM	OSCE - using the Calgary-Cambridge Observation Guide Checklist (SP and observer)	M106

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P66	RM	Self-assessment - Likert scale	M107
P67	RM	Empathy and Clarity Rating Scale (ECRS) in OSCE evaluation	M108
P68	RM	Comfort and competence of working with sexual and gender minorities survey	M109
P69	IM (A)	Portfolio-based assessment	M110
P70	RM	Wechsler Adult Intelligence Scale (WAIS-IV)	M111
P71	RM	OSCE - Standardized nurse (SN) rating on professional collaboration	M112
P72	RM	Focus groups	M113
P73	IM	OSCE - 4 stations (SP)	M114
P73	RM	5-minute examination - Diagnostic Accuracy	M115
P74	RM	The Essential Elements of Communication (EEC) - rating scale on an OSCE (SP)	M116
P75	RM	Socio-cultural domains of patient safety - OSCE (SP)	M117
P76	RM	Electronic 'Just in Time Medicine' (JIT) clinical evaluation exercize (CEX)	M118
P77	IM	Group Assessment using faculty opinion survey	M119
P78	IM (A)	Assessor Rating Scale	M120
P79	IM	360-degree feedback	M121
P80	RM	PRONE score	M122
P81	RM	PREP assessment (Physician Review Program) Test Battery	M123
P82	RM	Chart stimulated recall (CSR)	M124
P82	IM	SP's who rated students on a 5-point scale for communication skills	M125
P83	IM	Center for Personalized Education for Physicians (CPEP) multivariate competence assessment	M126
P84	RM	Patient-physician Interaction Quality Checklist	M127
P85	RM	Team Objective Structured Bedside Assessment (TOSBA)	M128
P86	IM	GMC Test of Competence (ToC) - including a written knowledge test and an OSCE	M129
P87	IM	GMC Test of Competence (ToC) - including a written knowledge test and an OSCE	M130
P88	RM	Knowledge-based questionnaire	M131
P89	RM	360-degree Assessment	M132
P90	IM	OSCE (SP)	M133
P91	RM	Emergency Medicine Milestones Survey	M134
P92	IM	Vignettes	M135
P92	IM	SP's	M136
P92	IM	Chart abstraction	M137
P93	IM	Referral for fitness to practice investigation	M138
P94	IM	OSCE	M139
P95	RM	The Jefferson Scale of Physician Empathy (JSE)	M140
P96	RM	The Scale of Attitudes Toward Disabled Persons (SADP)	M141
P97	RM	The Jefferson Scale of Physician Empathy (JSE)	M142
P98	RM	The Almost Perfect Scale-Revised (APS-R)	M143
P98	IM	Grade Point Average (GPA)	M144

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P99	RM	The cross-cultural care survey (CCCS)	M145
P100	IM	Workplace Based Assessments (WPBA) including: mini-clinical evaluation exercize (mini-CEX), case-based discussion (CbD), direct observation of procedural skills (DOPS), and mini-peer assessment tool (mini-PAT)	M146
P101	IM (A)	Workplace Based Assessments (WBA's)	M147
P102	IM	OSCE	M148
P103	RM	Support vector machine combined with an ordinal partitioning model (SVMOP)	M149
P104	RM	Professional Fulfilment Index (PFI)	M150
P105	RM	Epworth Sleepiness Scale	M151
P105	RM	QOL Burnout and depression	M152
P105	RM	Self-reported medical errors	M153

Notes.

EM: Existing Measures, RM: Research Methods, (A): Assessor assessment.

Papers in the manuscript are referenced in the format $^{P1,2,3,4,5\ etc.}$ individual measures are referred to as ^{M1,2,3,4,5} etc.

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MEASURING COMPETENCE IN DOCTORS

Supplemental Material File 4

List of Measures in Each Domain

Domain 1				Domain 2				Domain 3			
M1	M33	M72	M117	M2	M41	M88	M129	M2	M51	M100	M147
M2	M34	M73	M118	M6	M43	M90	M130	M4	M52	M102	
M6	M36	M75	M119	M8	M45	M93	M134	M5	M57	M103	
M8	M37	M82	M124	M9	M49	M95	M136	M6	M59	M104	
M9	M38	M83	M125	M13	M50	M97	M139	M8	M60	M106	
M10	M39	M85	M126	M14	M51	M98	M148	M9	M63	M109	
M11	M41	M86	M128	M19	M54	M99		M10	M65	M111	
M12	M43	M87	M129	M20	M55	M100		M13	M70	M114	
M13	M44	M88	M130	M22	M56	M102		M14	M72	M115	
M14	M45	M89	M131	M24	M57	M103		M15	M73	M118	
M17	M49	M90	M133	M25	M59	M104		M19	M74	M119	
M19	M50	M95	M134	M28	M60	M106		M28	M75	M126	
M21	M51	M97	M135	M29	M63	M114		M29	M77	M129	
M22	M54	M98	M136	M30	M65	M115		M31	M84	M130	
M24	M55	M99	M137	M32	M70	M117		M33	M87	M135	
M25	M57	M100	M139	M33	M72	M119		M34	M88	M136	
M26	M59	M102	M144	M34	M73	M122		M36	M90	M139	
M27	M60	M106	M146	M36	M77	M124		M37	M91	M141	
M28	M63	M109	M147	M37	M83	M125		M38	M97	M143	
M30	M65	M114	M148	M38	M85	M126		M43	M98	M145	
M32	M70	M115		M39	M86	M128		M45	M99	M146	

Domain 4			Domain 5			Domain 6		Domain 7		Domain 8	
M8	M67	M112	M1	M98	M139	M3	M75	M8	M75	M18	M94
M9	M68	M116	M22	M99	M148	M8	M79	M13	M76	M27	M96
M16	M83	M117	M33	M100	M150	M9	M80	M18	M90	M29	M103
M20	M84	M118	M36	M102		M10	M82	M19	M91	M31	M104
M25	M85	M119	M38	M103		M11	M89	M25	M103	M32	M113
M29	M86	M121	M41	M104		M21	M91	M27	M104	M33	M129
M33	M88	M125	M43	M106		M22	M99	M29	M113	M38	M130
M35	M92	M126	M44	M114		M23	M100	M31	M117	M43	M134
M36	M93	M127	M51	M115		M24	M103	M32	M119	M44	M139
M37	M95	M128	M52	M117		M26	M104	M33	M126	M47	M143
M38	M97	M129	M59	M119		M27	M111	M37	M129	M48	M150
M41	M98	M130	M60	M121		M29	M117	M38	M130	M51	M152
M43	M99	M132	M63	M125		M31	M118	M40	M133	M52	
M44	M100	M133	M65	M126		M32	M123	M43	M136	M56	
M46	M102	M136	M70	M128		M33	M128	M47	M139	M59	
M52	M103	M139	M72	M129		M34	M129	M51		M60	
M58	M104	M146	M83	M130		M37	M130	M52		M71	
M59	M106	M147	M85	M133		M38	M139	M53		M75	
M60	M108	M148	M90	M136		M41	M144	M59		M80	
M64	M109	M149	M95	M137		M43	M146	M60		M90	
M66	M111		M97	M138		M44	M147	M72		M91	

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Domain 9		Domain 10		Domain 11		Domain 12
M8	M96	M4	M121	M1	M138	M8
M12	M101	M5	M126	M4	M146	M18
M17	M105	M7	M129	M5	M147	M31
M18	M107	M8	M130	M8	M150	M48
M23	M113	M9	M132	M9	M153	M78
M25	M124	M14	M136	M14		M80
M30	M126	M15	M140	M15		M129
M33	M129	M36	M142	M23		M130
M43	M130	M43	M146	M36		M150
M52	M134	M46	M147	M43		M151
M56	M143	M58		M52		
M61		M59		M61		
M62		M60		M62		
M66		M61		M91		
M69		M62		M117		
M71		M64		M121		
M74		M84		M122		
M80		M91		M123		
M81		M106		M126		
M92		M108		M129		
M94		M109		M130		

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6-7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7-8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8-9

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Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8-9
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-15
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10-19
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	20-22
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	21
FUNDING			

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Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22

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